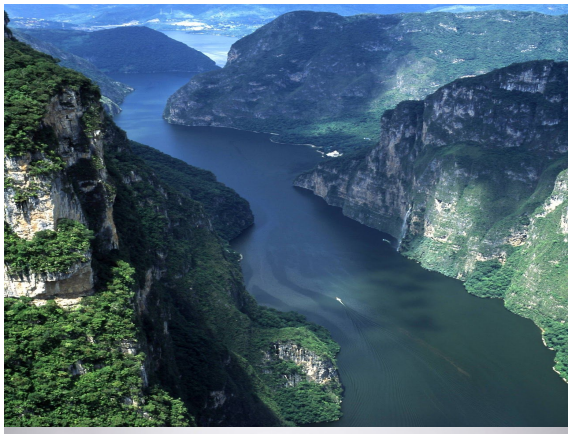
Trent University
Peterborough, Ontario, Canada

Watershed Management Plan for the Grijalva River Watershed in Tabasco and Chiapas, Mexico



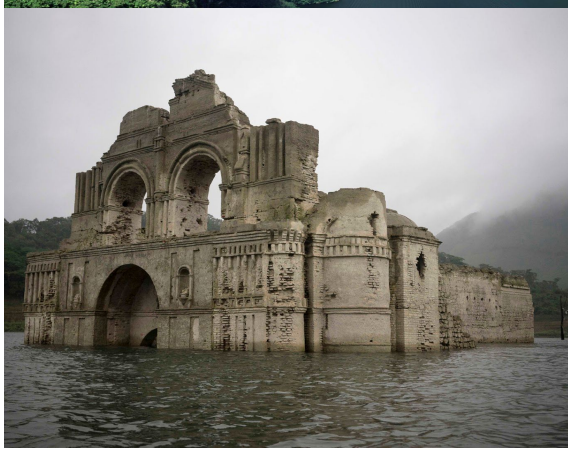


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Table Of Contents

1.0 - Executive Summary…....……...……..……...……1
2.0- Acknowledgements……..2
3.0 - Introduction……..……..……2
3.1 Geographical Area……..……...……?
3.2 Purpose of Study and Research Objective ……?
4.0- Background……...……..……..4
5.0- Approach……...……..……..4
6.0-Methods and Results…….…….…….……..4
6.0.1 Physical Characterization of Watershed Region……...…….……...……4
6.0.2 Boundary Delineation…….…….....……....4
6.0.3 Topography and Relief ……...……..……..4
6.0.4 Stream Morphometric relationships……...……...……?
6.0.5 Watershed Network Characterization ……..……...…..……...4
6.1-Land Resource Inventory……..……..…?
6.1.1 Geology…….…….……?
6.1.2 Soils…….…….…?
6.1.3 Vegetation…….…….…?
6.2 - Meteorological Stations and Climatic Variables.……...……..……...…?
6.2.1 Selection and Mapping of Meteorological Stations……...……...…?
6.2.2 Estimation of Parameters……..……...……..?
6.2.3 Mapping the Spatial Variability of Climatic Parameters……...……...…?
6.3 - Estimation of Runoff……..……..……..?
6.3.1 Introduction……..……..…….……?
6.3.2 Runoff Coefficients……...?
6.3.3 Runoff Results ……..?
6.4 - Land Use Inventory……...…….……..?
6.4.1 Agricultural Land Use……...……?
6.4.2 Present Land Use Types…….……..……...?
6.4.3 Potential Land Use Types ...……..……?
6.4.4 Land Utilization Types (LUT) Present and Potential……..……...?
6.5 Ecological Zones…….....?
6.6 Watershed Land Capability, Suitability, and Land-Use Planning……..…?
6.6.1 Land Capability Assessment……...?
6.6.2 Mapping of Environmental "Hotspots".....?
6.7 Land Use Planning and Conflict Resolution……..……...…..……?
6.7.1 Optimal Land Use Planning Scenario……..…...……..……?
6.8 Land Degradation Assessment…….……..?
6.8.1 Land Degradation Indicators……..……...?
6.7.2 Land Degradation Mapping……..…….……?
6.9 Water Quantity and Quality in the Watershed……..……...……....?
6.9.1 Design of a Monitoring Program for the Watershed……..……....?
7.0 - Discussion ……...…....……...……....?
8.0 - Bibliography……...……..……?
9.0 List of Appendices……...……..……?
9.1 Appendix A……...…….……?

List of Figures

1.0. Executive Summary

The purpose of this study was to develop a management plan that incorporates both the biophysical and human aspects of the Grijalva Watershed. A strategy was developed in order to manage these important components and execute them in order to conduct this study. The study was separated into five parts, in which we conducted research to find the characteristics of the watershed, defining the eco zones within the watershed, analyzing climatic inventory, estimating surface runoff, and assessing Land Suitability and Land Capability. Once completed, it was found that the Grijalva watershed is susceptible to climate change, such as severe weather events, that will have negative implications in the region.

2.0 Acknowledgements

We would first like to thank our professor, Doctor Raul Ponce-Hernandez of the Geography Department at Trent University. The door to Professor Ponce-Hernandez's office was always open whenever we ran into issues or had questions about our research. He considerably allowed us to do our own work, but steered us into the right direction whenever he thought we needed it.

We would also like to thank the TA's who were involved in assisting us during the research process: Kate Batelean and Adam Bird. Without their passionate participation and input, the research process would have not been successfully completed. We would also like to acknowledge the Mexican government as they provided us all the necessary resources and data to help kick start and complete our research. We are gratefully indebted to them and are proud to have completed this project in hopes of paving a brighter future for the Grijalva Watershed, and Mexico as well.

3.0. Introduction

The Grijalva is the largest river in Central America. Located in the country of Mexico, it drains a tremendous amount of water in the states of Chiapas and Tabasco (Lineart et al. 2011). The river is also home to one of Mexico's biggest watershed.Originating in Guatemala, the Grijalva watershed has an area of 87, 687 km2, consisting of 119 counties, with a population over 5.3 million (Lopez, 2009). It's region is classified as hot tropical climate The watershed's hydrological characteristics are driven by the two distinct seasons. These two seasons are the dry season which extends from November to May, and the wet season which starts in June and ends in October (Lineart et al. 2011).

The Grijavla Watershed is highly important to the geological and hydrological processes of Mexico. It contains approximately 27-40% of stored water of the country (Lopez, 2009). It is also important to the local residents, as the agriculture industry serves as major component to their GDP. Unfortunately, the watershed is at risk due to anthropogenic induced climate change. The region is at risk of increased severe weather events such as droughts and flooding. A study conducted by the Mexican government found that the watershed is susceptible to high extreme weather events, and is at risk of sea level rise (O'Hare, 2016). The effects of climate change are already being felt. In 2007, a flood in the region affected 75% of the state of Tabasco's population, and damaged 30% of the state's GDP (O'Hare). O'Hare (2006) mentions how the first decade of the 21st century has impacted 3 million people of the region, and has resulted in losses of close to US 5 billion dollars due to a reduction of agricultural productivity.

3.1 Geographical Area

The Grijalva watershed basin is 87, 687 km2 in size. Its headstreams, the largest of which is the Cuilco, rises in the Sierra Madre of Guatemala and the Sierra de Soconusco of Mexico. The Grijalva flows Northwest through Chiapas state, where it is known locally as the Río Grande de Chiapa, or the Río Chiapa. After leaving a lake created by the Malpaso Dam, it turns northward and eastward, roughly paralleling the Chiapas-Tabasco state border. Veering northward again at Villahermosa, the capital of Tabasco, it receives the main length of the Usumacinta River and empties into the Gulf of Mexico and the Bay of Campeche, 10 km north-northwest of Frontera. The river is navigable by shallow-draft vessels for approximately 95 km upstream from the bay and for several stretches along its middle and upper course. The Grijalva River's total length is approximately 480km

3.2 Purpose of Study and Research Objective

The Purpose of this document is to develop a management plan that encompasses, both, the biophysical attributes and human aspects of the Grijalva watershed and develops a strategy for managing important components. The research objectives of the management plan for the Grijalva watershed are to identify best uses of specific land types, reduce the risk of flooding, and protect the overall quality of the water. To accomplish these objectives a variety of data sources are used as well as a collection of statistical analyses. Climatic data of the watershed was analyzed, surface runoff estimations were developed, and land suitability and capability matrices were used.

4.0 Background

The Grijalva watershed is one of extremely large spatial boundaries. For the purpose of time and project management, the watershed was discontinued in areas wherein the actually watershed boundary would have extended slightly further. This information is important when considering total inputs into the hydrologic system and current land use impacts in the area that was not included in the watershed management plan. The impacts on the hydrologic system could, therefore, be slightly more dynamic than that which are predicted in this watershed management plan.

5.0 Approach

For the purpose of this management plan, the thematic approach was employed. This is a theme focused approach wherein thematic maps are used to divide resource variability into ecological zones. To do this, data is inventoried and mapped out by ecological theme. In this management plan, the following categories are mapped using geographic information systems; soils, land use, climate, geology, hydrology, and land suitability. The thematic approach was the method of choice because of the type of data that was available from the states encompassed by the Grijalva watershed. Also, the thematic approach reveals more detail and is more economically feasible in terms of budget.

6.0 Methods and Results

6.0.1 Physical Characterization of Watershed Region

The physical characterization of the watershed was completed using a 250, 000 scale topographic map under tracing paper obtained from Trent University. A light table was used to project the analog map image through the tracing paper in order to create the watershed boundary along with stream flow data and locate a discharge point. Hydrologic networks were drawn within the watershed boundary including lakes and reservoirs. To digitize the watershed, the four pieces of tracing paper were connected and a photo was taken with a high resolution camera. The photo was uploaded and imported to a Geographic Information System program called ArcGIS. Once complete, the watershed was digitized. The watershed's boundary, streams, stream order, ecozones,

6.0.2 Boundary Delineation

The uploaded image of the hand-drawn map was opened in a tiff-file format. The image was then opened in ArcGIS and the control points used for georeferencing. The watershed boundaries were identified by understanding the direction of drainage from elevated areas of the landscape. Satellite images were consulted for areas wherein the topography was unknown after tracing occurred.

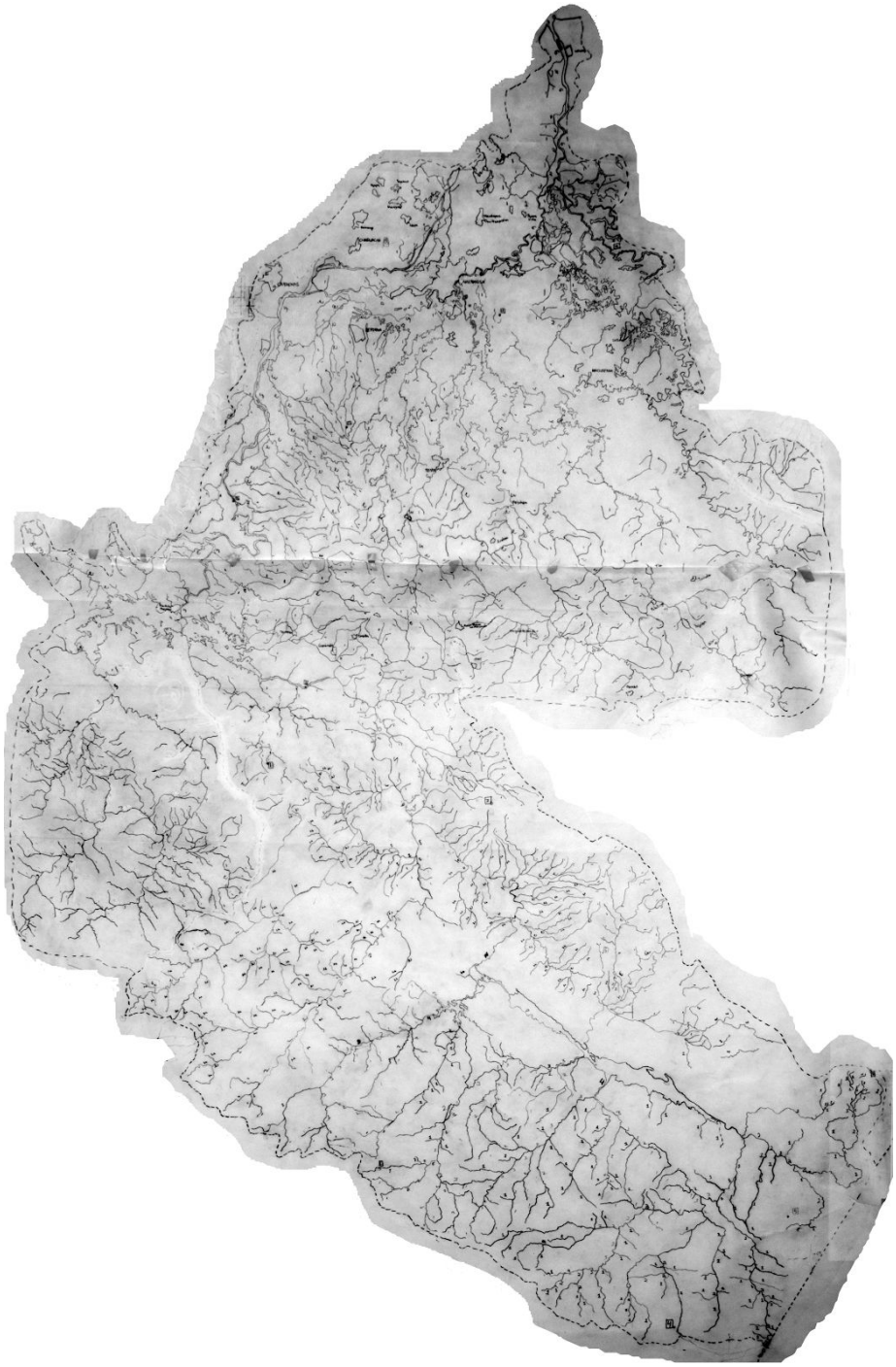

Figure 1. Map depicting the entire Grijalva Watershed, sub-watersheds, hydrological network, and stream orders.

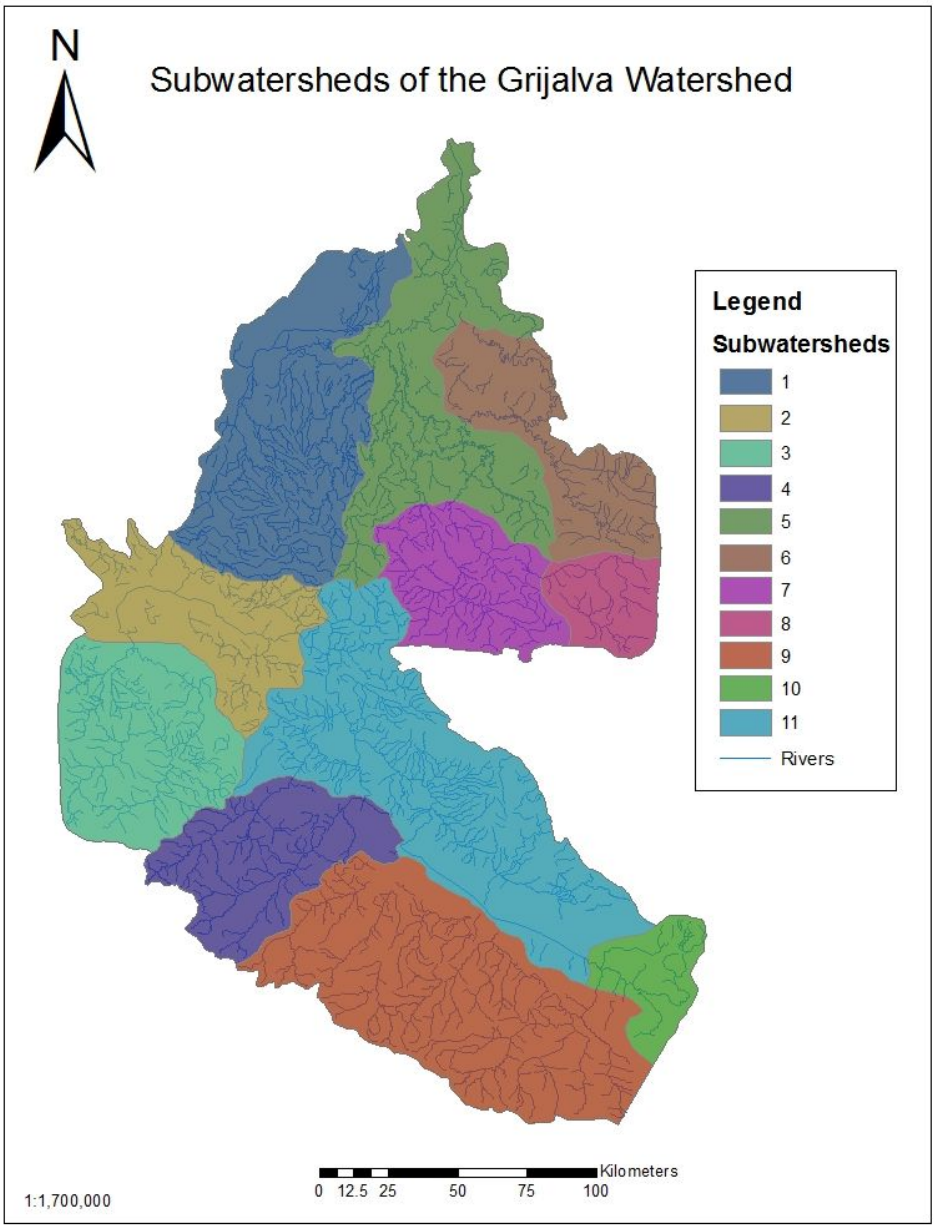

Figure 2. Map of hydrological subwatershed network in Grijalva watershed.

6.0.3 Topography and Relief

The highest point in the Grijalva watershed is 3001 meters above sea-level which is located on the northern end of the watershed. The watershed is characterized by having unrestricted freshwater input which, is then, heavily influenced by topography and relief (Reyes, et al., 2004). Topography and relief are very important when characterizing the physical

watershed as they were the main characteristics consulted for boundary delineation. Relief is defined by the highest and lowest points in a given area. This is one of the most important parts in understanding a watershed and its boundaries as the flow of water is constrained by the relief. Topography includes features such as; elevation, cross-sections, slope gradient, and stream drainage patterns which are obeyed by the hydrology.

6.0.4 Stream Morphometric Relationships

Delineation of the watershed boundary stream were all ordered. It was determined that a total of 7 streams orders present in the watershed with 1,423 streams in total. The below table categorizes these 1,423 streams into their respective stream orders. As can be seen in Figure 1 the discharge point is at the northernmost point of the watershed with the largest stream orders.

Stream Order	Numbers of Streams
1	1109
2	237
3	52
4	15
5	4
6	5
7	1

Table 1. The number of streams of each order in the Grijalva hydrological network

6.0.5 Watershed Network Characterization

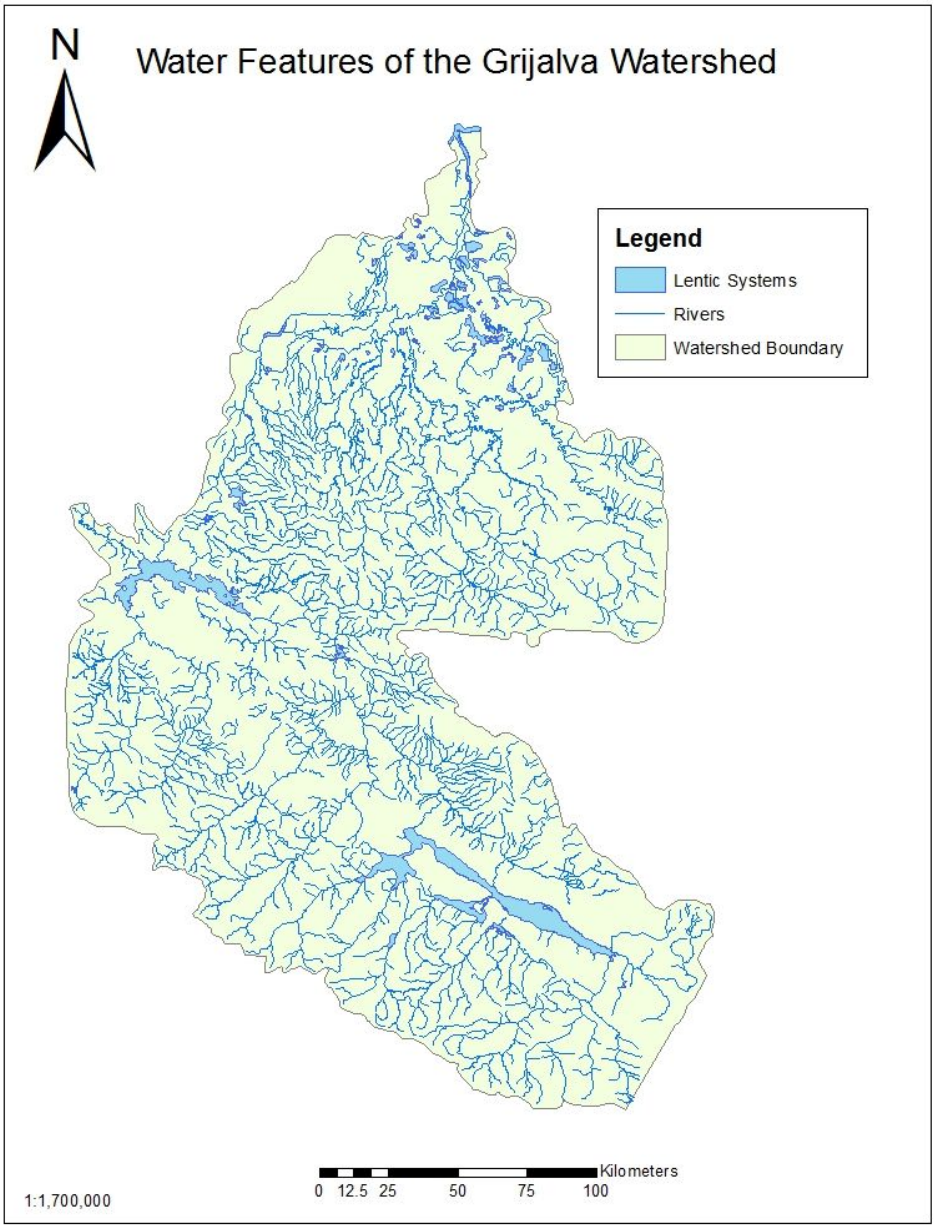

Figure 3. Map illustrating the water features within the watershed including; rivers and lentic systems.

To characterize the stream networks, the team started from the largest order stream which, is the Grijalva River and worked outwards. We then, ordered the streams based on the number of streams adjoining to them. A first order stream is one that does not have another stream connected to it while a second-order stream is where two first-order stream meets, this procedure continues until all of the stream are ordered. The largest order stream in the Grijalva watershed is a seventh-order stream which is the river itself.

As can be seen below, Figure _ through _ illustrates the correlation between stream order and stream length as well as the correlation between between stream order and the number of streams given order. These relationships were determined for the entire watershed and suggest that there are underlying factors that preside over the structure of the different stream attributes in a similar and predictable manner (Ponce, 2017).

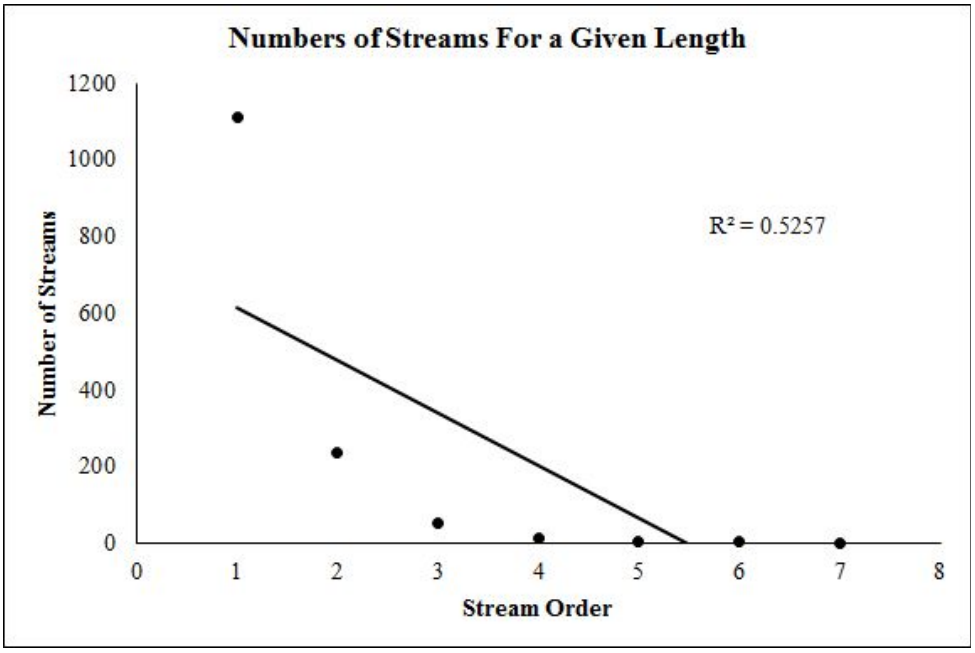

Figure 4. Negative relationship between stream order and the number of given streams of a given order.

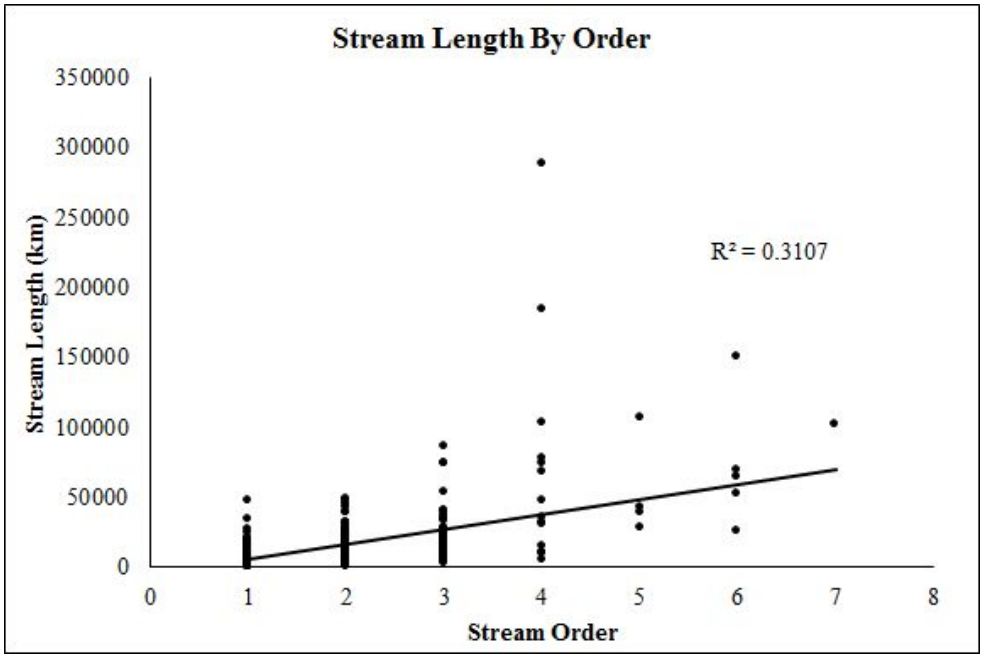

Figure 5. Positive relationship between stream order and stream length.

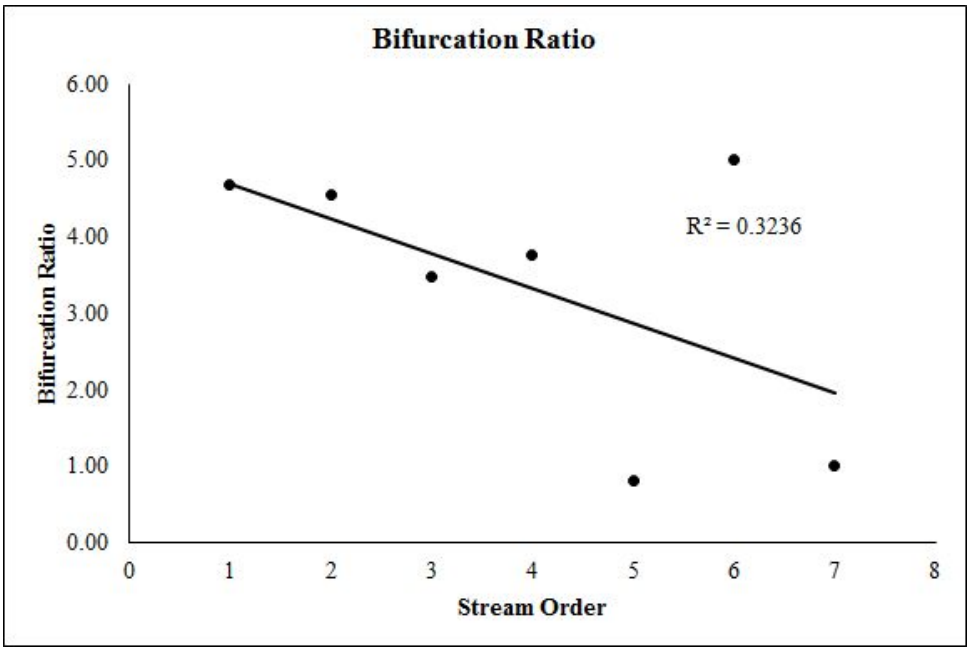

Figure 6. Negative correlation between Bifurcation Ratio and stream order, with a R^2= 0.3236.

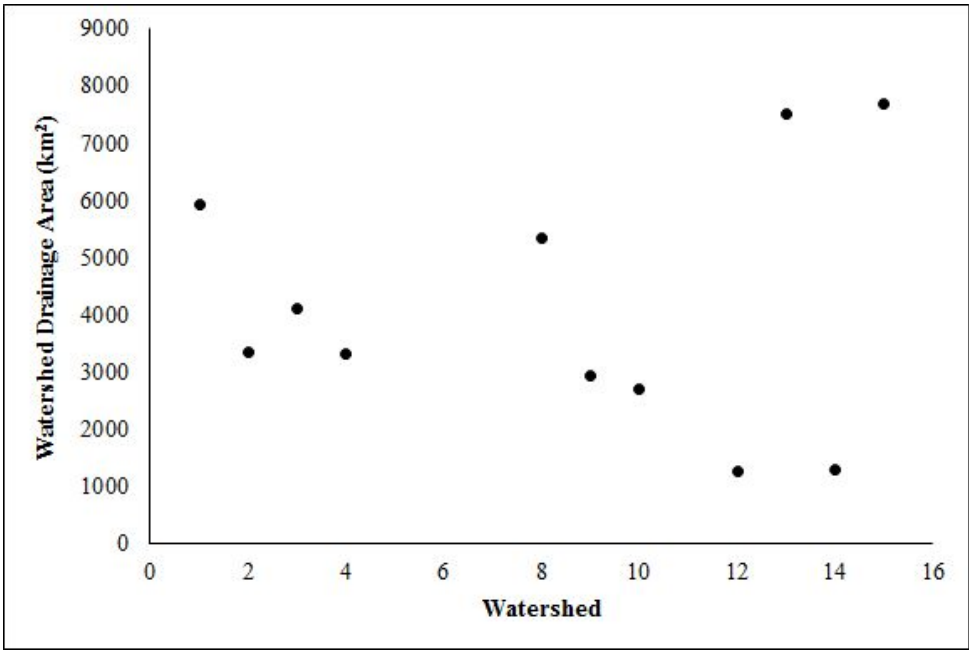

Figure 7. Watershed Drainage Area had no correlation based on the number of watershed.

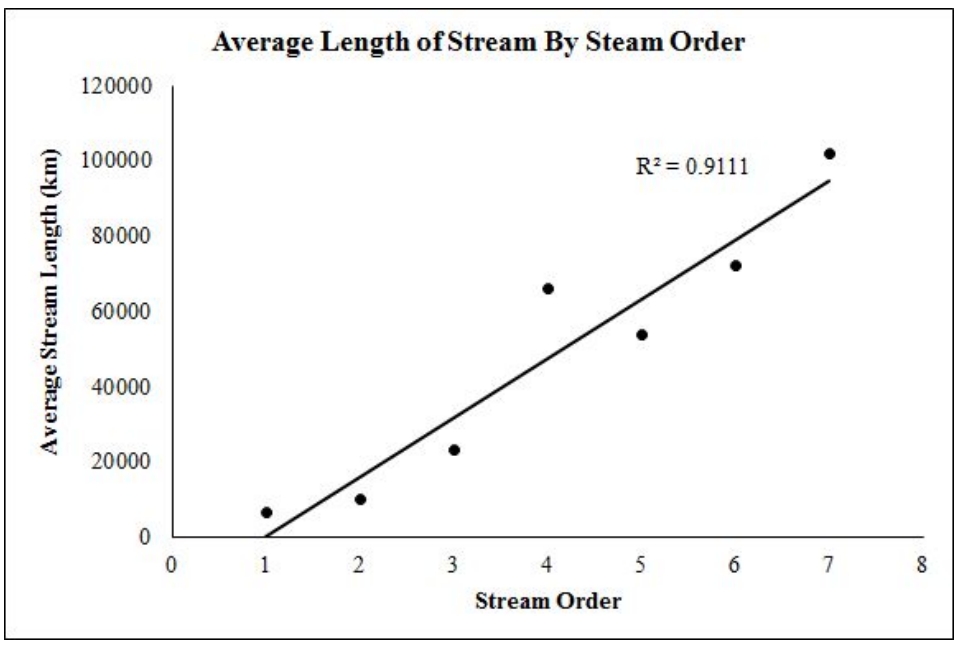

Figure 8. Positive relationship between Average Stream Length and Stream Order with R^2=0.9111.

6.1 Land Resource Inventory

6.1.1 Geology

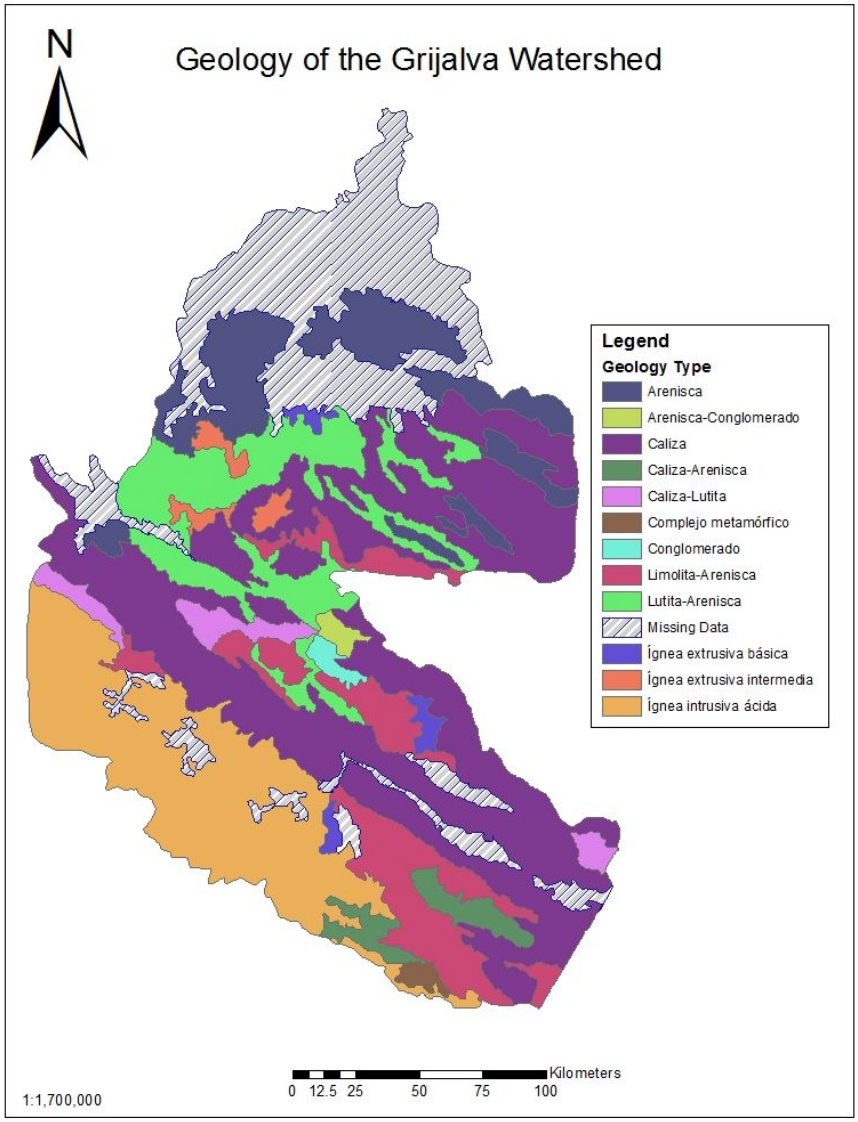

Figure 9. Geology of the Grijalva Watershed

The largest geology classes found in the Grijalva watershed were that of Caliza (Limestone) and ignea intrusiva acida (intrusive igneous rock). There was also a large portion that was missing data or not interpolated. Missing Mexican data prevented the completion of a full geology data set for the Grijalva watershed.

6.1.2 Soils

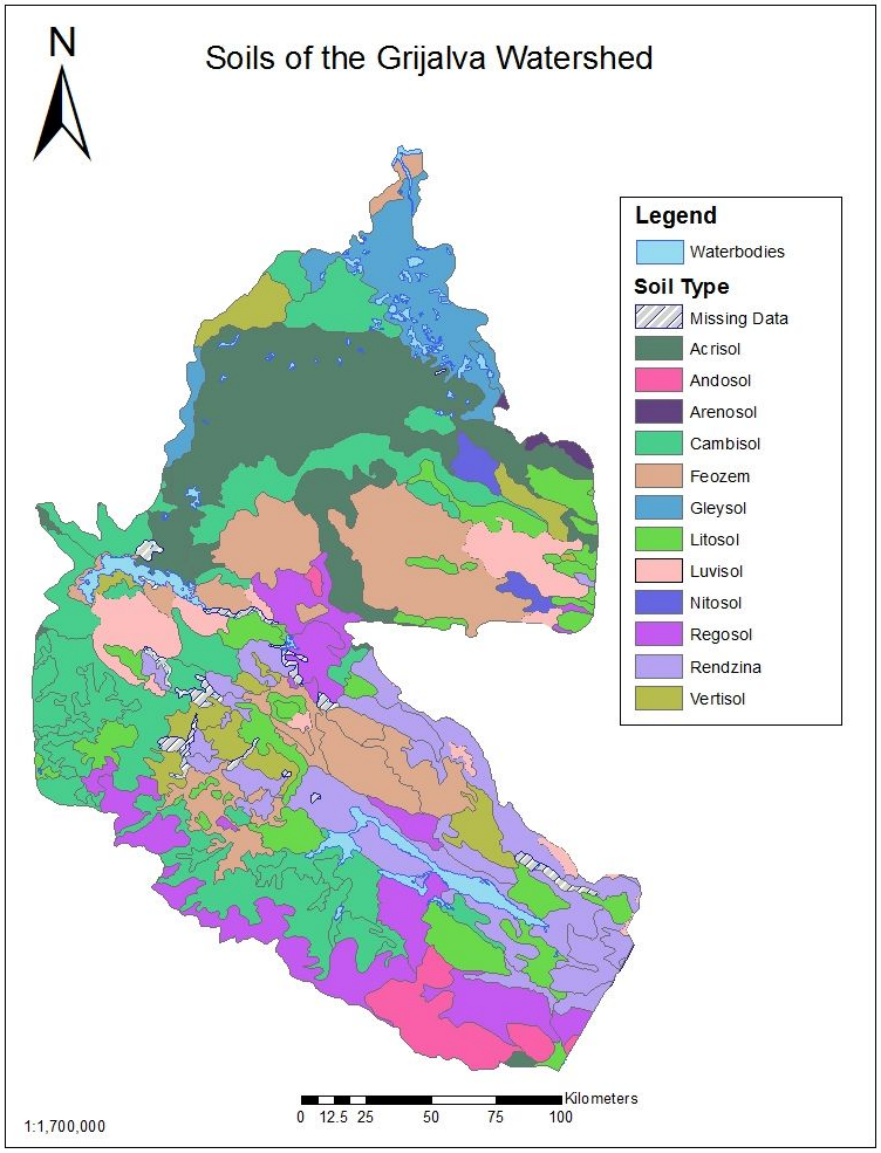

Figure 10. Soil classification in the Grijalva Watershed

The most dominant soil types found in the grijalva watershed were cambisol and acrisol found in the north west of the boundary. Once again there were small areas with missing data affecting the total number of soils. The large watershed area led to the vast number of soil types for data collection.

6.1.3 Vegetation

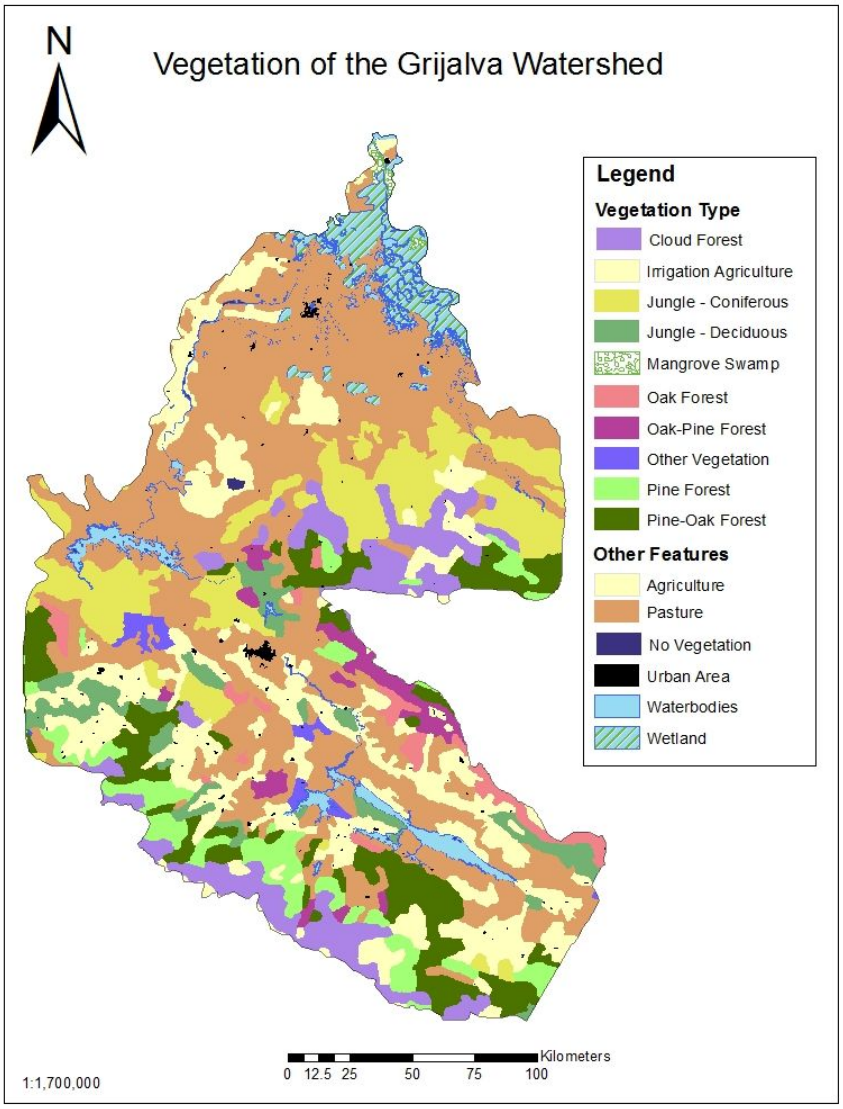

Figure 11. Vegetation in the Grijalva watershed

Vegetation was primarily dominated by pasture land followed up by coniferous jungle. The was a very small amount of urban areas compared to other vegetations. The largest urban areas are surrounded by pasture land which covers most of the north and down into the south. There is high diversity between vegetation types along the lower portion of the watershed.

6.2 Meteorological Stations and Climatic Variables

6.2.1 Selection and Mapping of Meteorological Stations

In order to build climatology for the region, we took 10 years of continuous data from weather stations in the watershed. Daily data was collected from 17 weather stations, over a course of 10 continuous years. We used the time frame of 2001 to 2010 in order to have the most recent data as well as avoid certain inconsistencies in the data that will be discussed later.

Appendix A (Tables A.1-A.18) holds the weather station data and their monthly average values used for the watershed. These tables were created using Pivot tables of the daily data. Figure XOX shows the spatial distribution of the weather stations.

<u>Note on Distribution:</u> There was an initial pool of 28 weather stations for proper distribution with data available for both temperature and rainfall. However, due to inconsistency in the time frame, the pool dropped to the resulting distribution of 16 weather stations as seen in Figure XOX.

Distribution of Weather Stations Used

Figure 12. Meteorological stations used in this study.

6.2.2. Estimation of Parameters

The monthly averages from the weather stations was then averaged over the 10 years. Using the 10 year averages, we get monthly values for Temperature and Precipitation for each weather station. This step causes us to lose the resolution to witness 10 year weather patterns. However, using the averages of over 10 years for each month, we were able to plot annual climate patterns for each weather station. Table A.19 in Appendix A, summarizes the weather station data over a year averaged from the 10 year data.

The parameter of Potential Evapotranspiration (ETo) was calculated using the ETo Calculator (Food and Agriculture Organisation) developed by the FAO. Using the location and existing weather data, values for ETo in mm/day were obtained and consequently employed.

This climatic inventory for the weather stations, was plotted under the following parameters to create Length of Growing Period (LGP): (a) Average temperature > 6.5 degrees C, (b) Average Precipitation > 0.5 Potential Evapotranspiration.
Graphs in Appendix A demonstrate the obtained monthly values for each weather station.

The watershed displays a wide range of weather patterns, as demonstrated in Figure 13 and 14 below:

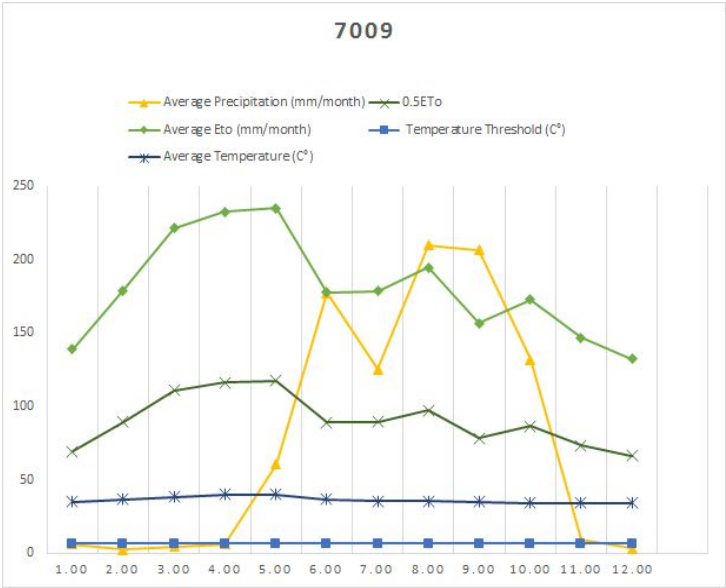

Figure 13. Climograph for meteorological station 7009 in the Grijalva watershed.

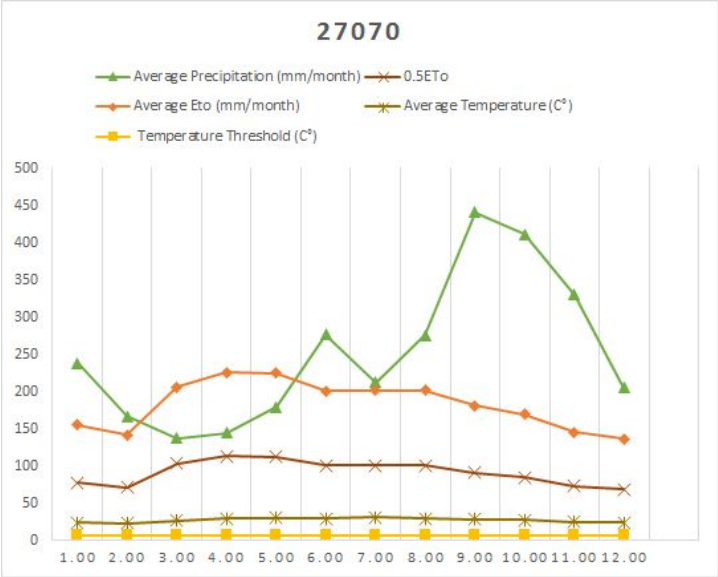

Figure 14. Climograph for meteorological station 27070 in the Grijalva watershed.

Weather Station 7009 showed the smallest LGP of just 157 days and, 27070 and 27042 showed the longest; 365 day LGP. However, across the variability, the watershed did not have the average temperature fall below the temperature threshold of 6.5 degrees C when the Precipitation was above 0.5 potential Evapotranspiration. Hence, the LGP across the watershed is solely precipitation dependant.

6.2.3 Mapping the Spatial Variability of Climatic Parameters

The data was then interpolated using Tension Spline. Figure XOX to XOX show the interpolation of data. The Spatial Pattern of the Average precipitation is very similar to the LGP, demonstrating the precipitation dependence of LGP on the average precipitation, under our prescribed pre-conditions.

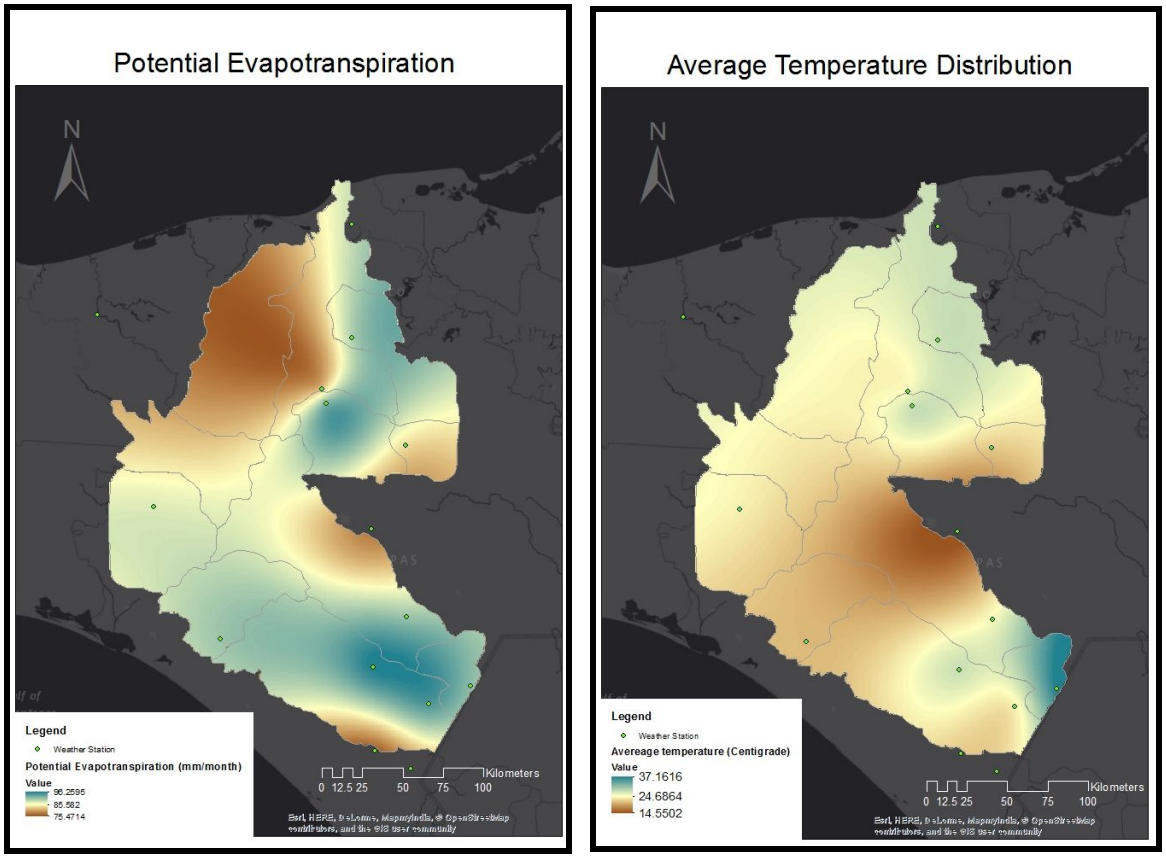

Figure 15. Potential evapotranspiration and average temperature distribution of the Grijalva watershed.

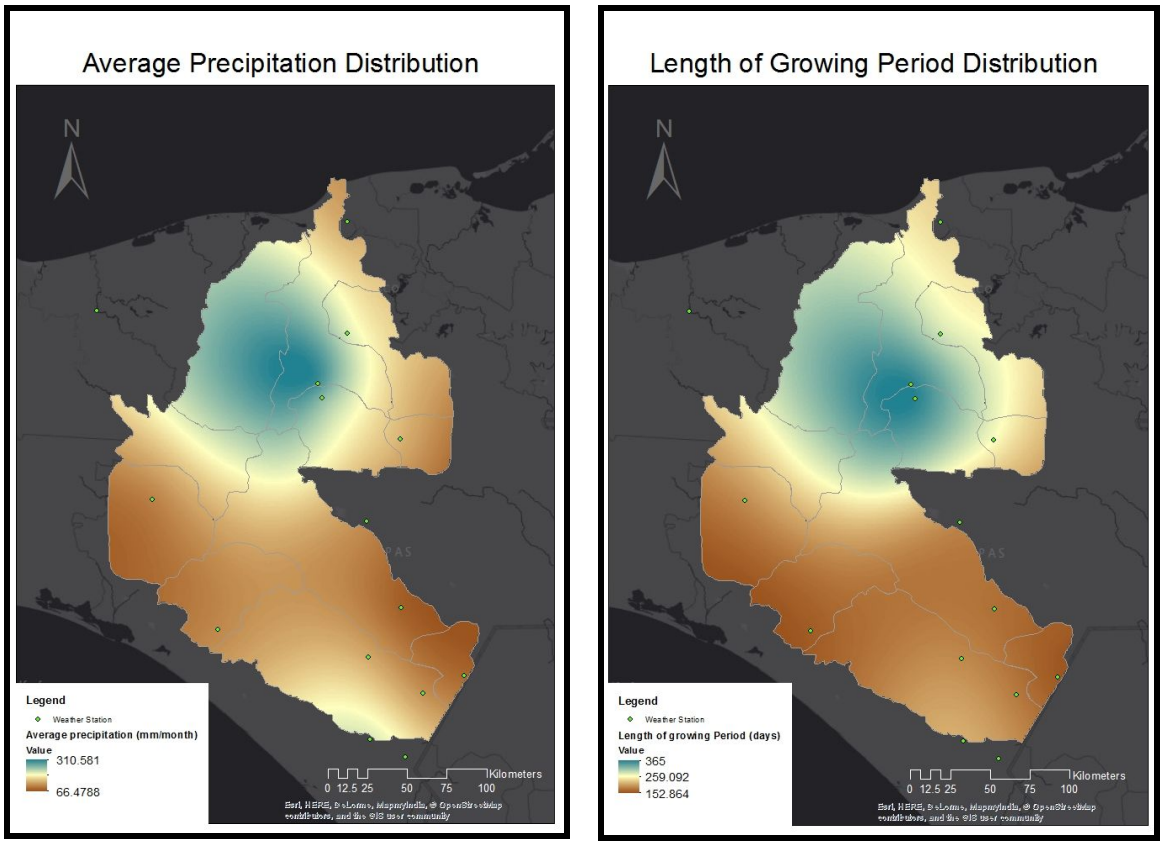

Figure 16. Average Precipitation and average length of growing period of the Grijalva watershed.

The final summary of the data, is as follows:

Table 2: Summary of climate data for all weather stations utilized.

STATION	Average ETo (mm/month)	Average Precipitation (mm)	Average Temperature (Degrees C)	Average LGP (days)
4024	92.5625	148.212	25.55929	238
7009	91.03167	86.06777	35.27547	**157**
7021	95.94458	132.9899	26.74658	190
7036	77.93508	114.4778	14.06111	190
7054	75.92208	206.4711	21.6532	214
27070	91.14208	251.4535	26.91616	**365**
7070	93.82125	117.1158	23.98431	180
7071	84.135	141.2571	22.01831	260
7094	90.02375	79.54159	24.87098	173
7333	63.23398	104.7878	18.23765	193

7349	90.42792	109.5798	21.25959	159
7376	87.40083	105.1666	23.69585	192
27050	88.22667	130.7261	26.548	238
27030	88.86288	203.6627	26.88076	282
27042	81.63125	290.745	25.2408	**365**
30167	90.45917	200.01	26.24271	276

Due to the proximity of some of the data, especially in the middle of the watershed, the LGP had to be manually limited within 365 days, since the mathematically interpolated values were exceeding the said, due to the proximity of the weather station data. The values were manually compared to the values in the tables to ensure the limits were correct. Station 27070 contributed the 365 day LGP.

Henceforth, is the analysis of our findings:

Inferences:

(a) The watershed has a varying amount of precipitation values that limit the LGP in the regions.
(b) Topography and precipitation show a strong relation, especially, in the mountainous region in the South West of the watershed. Orographic rainfall pattern is suspected.
(c) The North Segment of the watershed has longer LGPs, concentrated near the center. Due to the observed shape of the Average Precipitation distribution and related LGP, multiple factors can be said to influence the climate, including the presence the coastal regions and waterbodies within the watershed.

Limitations:

(a) The data has been summarized over multiple stages of averages from the initial daily data. Variability in weather patterns is unaccounted for.
(b) The resultant pool of weather stations used is not uniformly distributed. Many weather stations showed periods of inactivity after large observed precipitation values and the 31st of December of certain years. We suspect damage and scheduled maintenance, respectively, to be the cause. There were also certain years, like 1988, 2003, 2009 etc, when many of the weather stations stopped recording values and have done so since then. Hence, the number of usable weather stations shrunk considerably, forcing us to rely on spatial interpolation to generate values for large spatial stretches of the watershed. This limits the resolution of the data in different regions. For example, while it is more detailed near the center, it is very coarse in the North-East part of the watershed.
(c) The ETo values lacked parameters like sunlight, wind, slope etc. Hence, the value for ETo are limiting in precision and while are safe to use over data as summarised and relational as ours, it's values should not be used for any other interpretations.
(d) Soil moisture was neither calculated nor accounted for in the ETo or LGP. Sources of groundwater saturation or storage have not been accounted for.

6.3 Estimation of Runoff

6.3.1 Introduction

Surface runoff estimation was performed for the sub-watersheds of the Grijalva. The calculations of surface runoff is an important component of a mass water balance. The creation of a mass water balance looks to determine the volume of water flow through the stream network, in order to anticipate both flood and drought conditions that may be found through the hydrological system. Various equations along with IDF curves which used data from: stream network, sub-watersheds, stream length, sub-watershed slope all used to calculate surface runoff. Rainfall data was collected from local meteorological stations. Runoff coefficients are functions of land cover, slope, soil type, all calculated using the provided table. Calculations were completed for concentration time and maximum surface runoff for the sub watersheds of the Grijalva watershed.

6.3.2 Runoff coefficients

Runoff coefficients were determined for each sub-watershed, using the table below (Table 3) of coefficients and compiled land cover, slope percentage, and soil data.

Land Cover/ Slope	SOIL	TEXTURE	
	Loamy Sand & Sand	Clay Loan & Silty Loam	Clay
FOREST:			
O – 5%	0.10	0.30	0.40
5 - 10%	0.25	0.35	0.50
10 – 30%	0.30	0.50	0.60
>30%	0.30	0.50	0.60
GRASSLAND			
0 - 15%	0.10	0.30	0.40
5 – 10%	0.16	0.36	0.55
10 – 30%	0.22	0.42	0.60
>30%	0.22	0.42	0.60
CROPLAND			
O – 5%	0.30	0.50	0.60
5 - 10%	0.40	0.60	0.70
10 – 30%	0.52	0.72	0.82
>30%	0.52	0.72	0.82

Table 3. Runoff coefficient classification table

Concentration Time
Concentration time (Tc), or the amount of time a raindrop would take to run from headwaters to the discharge point of the the watershed, calculated using:

$$T_c = 0.02 \frac{L^{1.15}}{H^{0.385}}$$

Tc = Concentration time, L= length of the subwatershed, and H= height or elevation difference between the headwaters and discharge point.

6.3.3 Runoff results

A table of maximum runoff was created using the above methods to illustrate runoff characteristics present in the Grijalva watershed. The results were compiled and used in ARCGIS to create Figure 18. Variation between sub watershed maximum runoff is a result of differing slope, land cover, soil, and amount of rainfall received. Runoff values from low (light purple) to high (dark purple) in Figure 18.

Table 4. Maximum Runoff of each subwatershed in the Grijalva watershed.

Subwatershed	Sub-Watershed Area (km2)	Runoff Coefficient	Maximum Rainfall Intensity (mm/hr)	Maximum Runoff (m3)
1	59300.6	0.4	8	5313.3
2	33472.1	0.6	21	10824.9
3	40992.4	0.6	21	13256.9
4	33062.6	0.3	12	3332.7
8	53331.9	0.4	7	4181.2
9	29368	0.4	11	3618.1
10	27163.4	0.8	30	18710.2
12	12610.1	0.6	60	12711
13	75222.7	0.5	10	10531.2
14	12928.7	0.4	2	289.6
15	77006.9	0.6	20	23718.1

Maximum Runoff

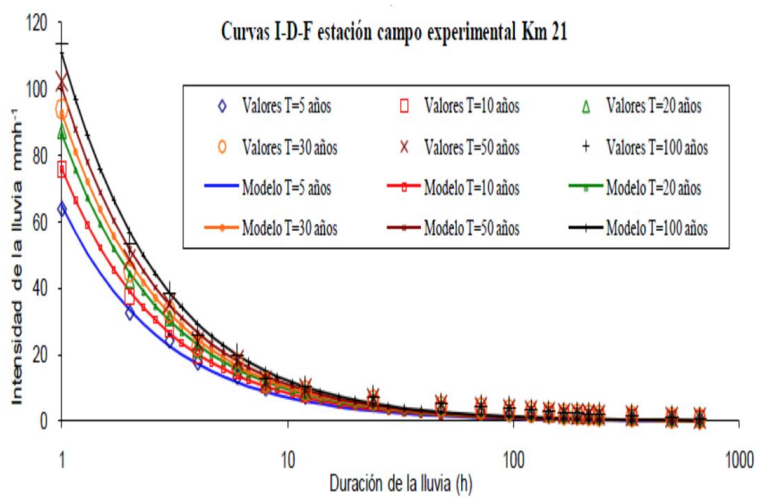

Figure 17. Shows the experimental IDF curve used to evaluate runoff

The Intensity-Duration-Frequency (IDF) curve found below was used as the best representation of the entire watershed. Concentration Time (Tc)/ Duration (d), were used as basis points for IDF correlation of obtained rainfall intensity (i). Duration used was 5, 30, and 50 years on the IDF. This was determined through the comparison of climate data peak values over 10 years with the peak values for the 10 year curves in IDW models. Similar values and trends were manually compared in order to ascertain the best representation of our watershed climatology for the 10 year period. The selected model was then used to infer Rainfall intensity for the different time frames and across the subwatershed concentration times, calculated in the previous step.

Henceforth is the equation used to calculate the maximum runoff for each watershed. The Rainfall intensity was inferred from the IDW curve, the Watershed Area from the digital polygon on the map.

$$Q = 0.028\ C \cdot i \cdot A$$

Q = maximum runoff (m^3/sec)
i = Rainfall intensity for a given return period (cm/hr)
A = Watershed Area (Ecozone or Hydrological Response Unit) (ha)
C = Runoff Coefficient

Concentration time (Tc) = Duration (d), and Frequency (f) = return period

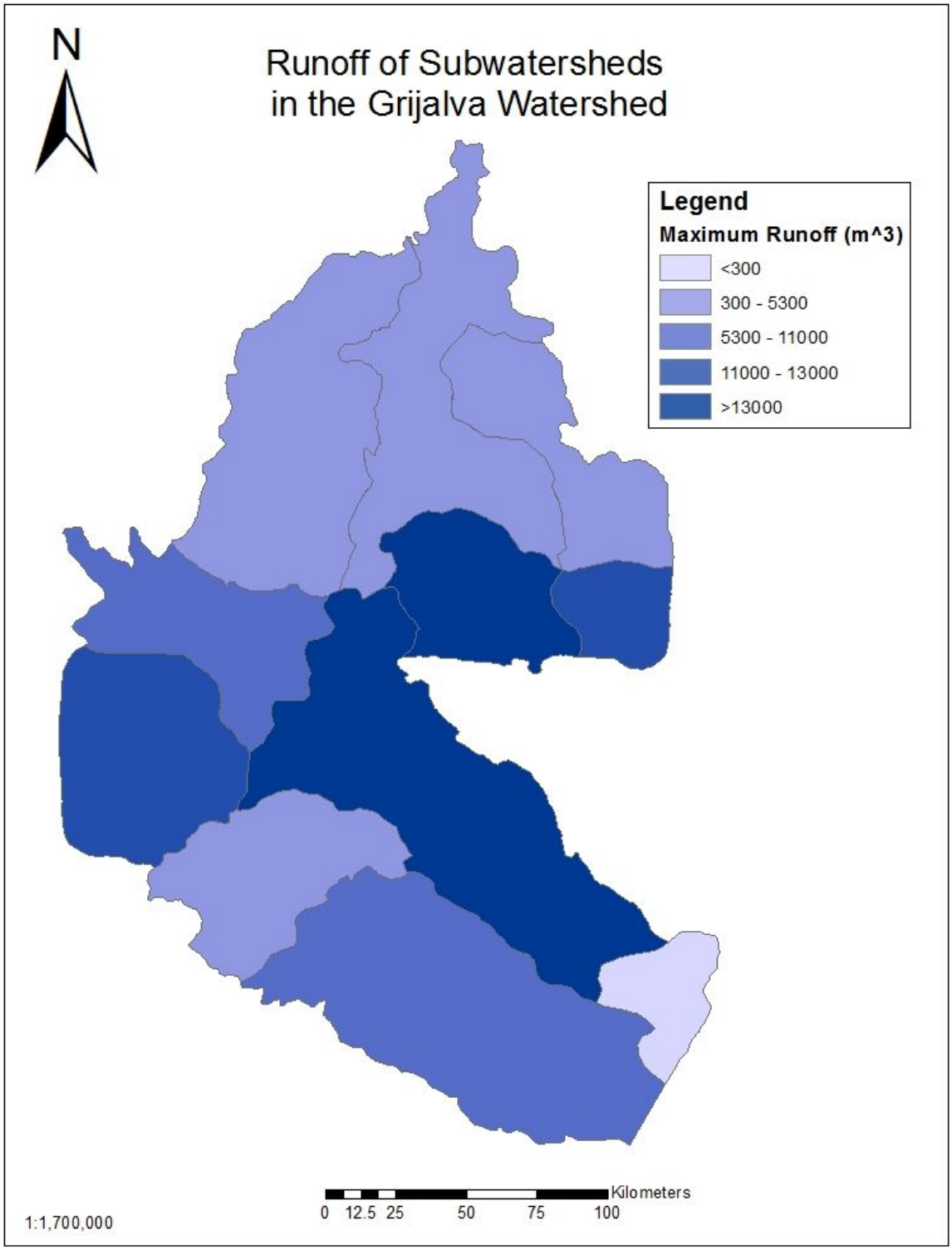

Figure 18 (a). Map of runoff values (m^3) per sub watershed in the Grijalva watershed.

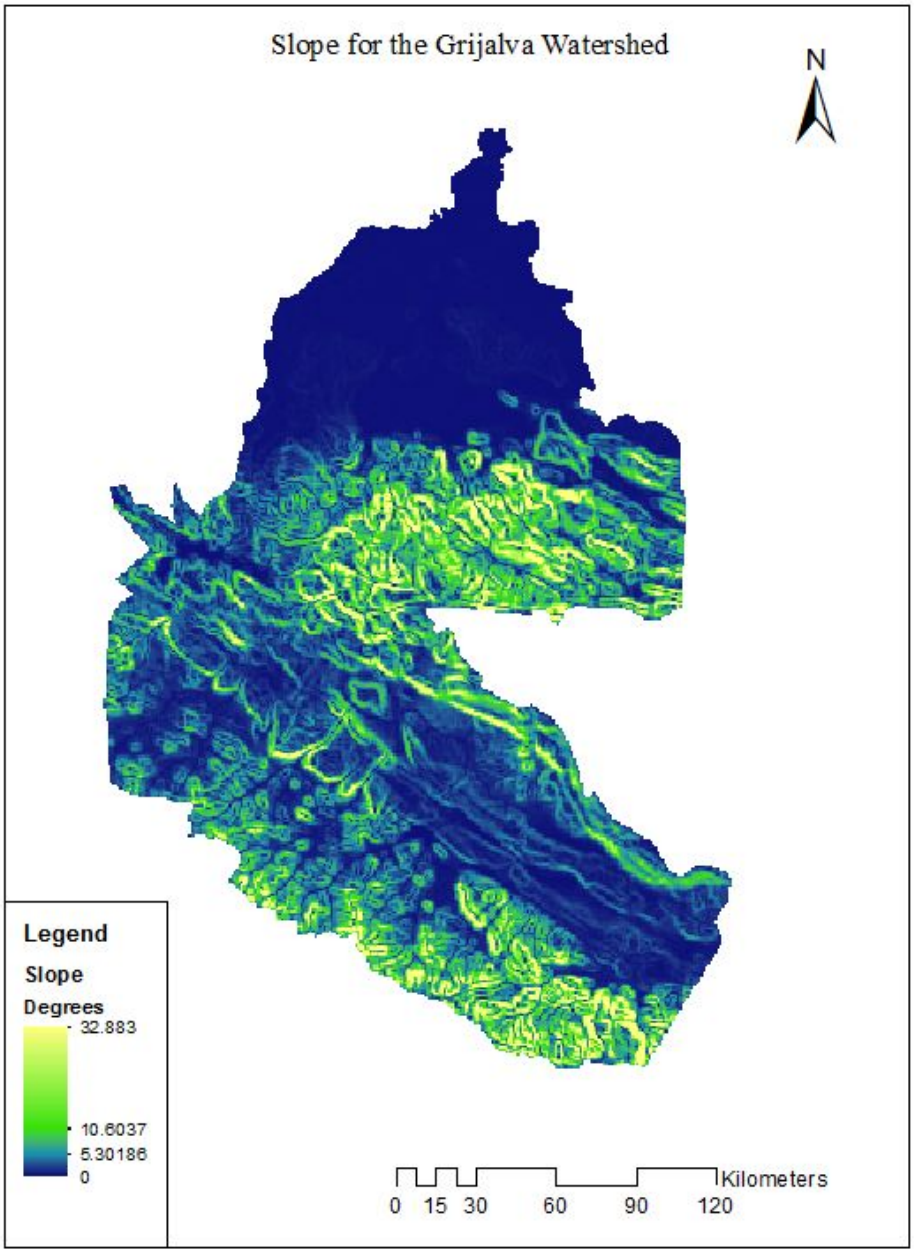

Figure 18(b): Slope Distribution of the Grijalva Watershed.

Inferences: Figure 18 shows the distribution of maximum runoff in the watershed. The divide between the Usumacinta and grijalva watersheds demonstrates the largest runoff value. On the contrary, the runoff is low and quite uniform in the North of the watershed. A comparison with the slope in Figure 18(b) shows a positive correlation between runoff and slope. A note must be made to the approach used. Elevation and length were used as the base parameters to determine the watershed maximum runoff values. It is a more indirect approach and has its limitations due to the said.

6.4 Land Use Inventory

This management plan analyzes the use of the current land and determines whether the land is being used appropriately based on capability factors. In order to understand the current state of the watershed, a land use inventory must be initiated. The land use inventory consists of information regarding current uses of land. To gather the data we used general land use information which was provided to us by the Mexican government.

6.4.1 Agricultural Land Use

As demonstrated in figure 19, the Grijalva watershed has approximately one third of its land designated as agricultural and pastoral use. This is a considerable portion of the watershed and is highly susceptible to degradation due to increased precipitation and flooding. There have been a multitude of studies on the vulnerability of crops in the Grijalva watershed, particularly, rainfed maize, and coffee production. The land use inventory exemplifies that this section of the country will be severely impacted by the anticipated extreme events caused by climate change (Lopez, 2009).

6.4.2 Present Land Use Types

According to the thematic map generated, the second largest portion of land is taken up by forests. There was limited data on forest types so it is difficult to assume the level of forestry that is occurring in the watershed. A section of natural land is designated as a wildlife preserve which goes by the official name of Reserva de la Pantanos de la Centla. This area is mostly wetland and would not be included in the possible areas for forestry to occur. Deforestation, however, is ranked as 9th in the world in Mexico which could be partially the responsibility of the forestry industry (Kolb & Galicia, 2012). Over 90 percent of production in the Mexico forestry industry comes from temperate forests. Chiapas as well as six other states in the country is listed as one of the main states contributing to the forestry industry. The state of Chiapas is also responsible for the remaining 10% of forestry occurring in tropical regions (Mexico-Forestry, 2017). Although we do not have data regarding exact areas wherein forestry is occurring, we know that forestry is a main industry in this area and that parts of the Grijalva watershed is being used for it.

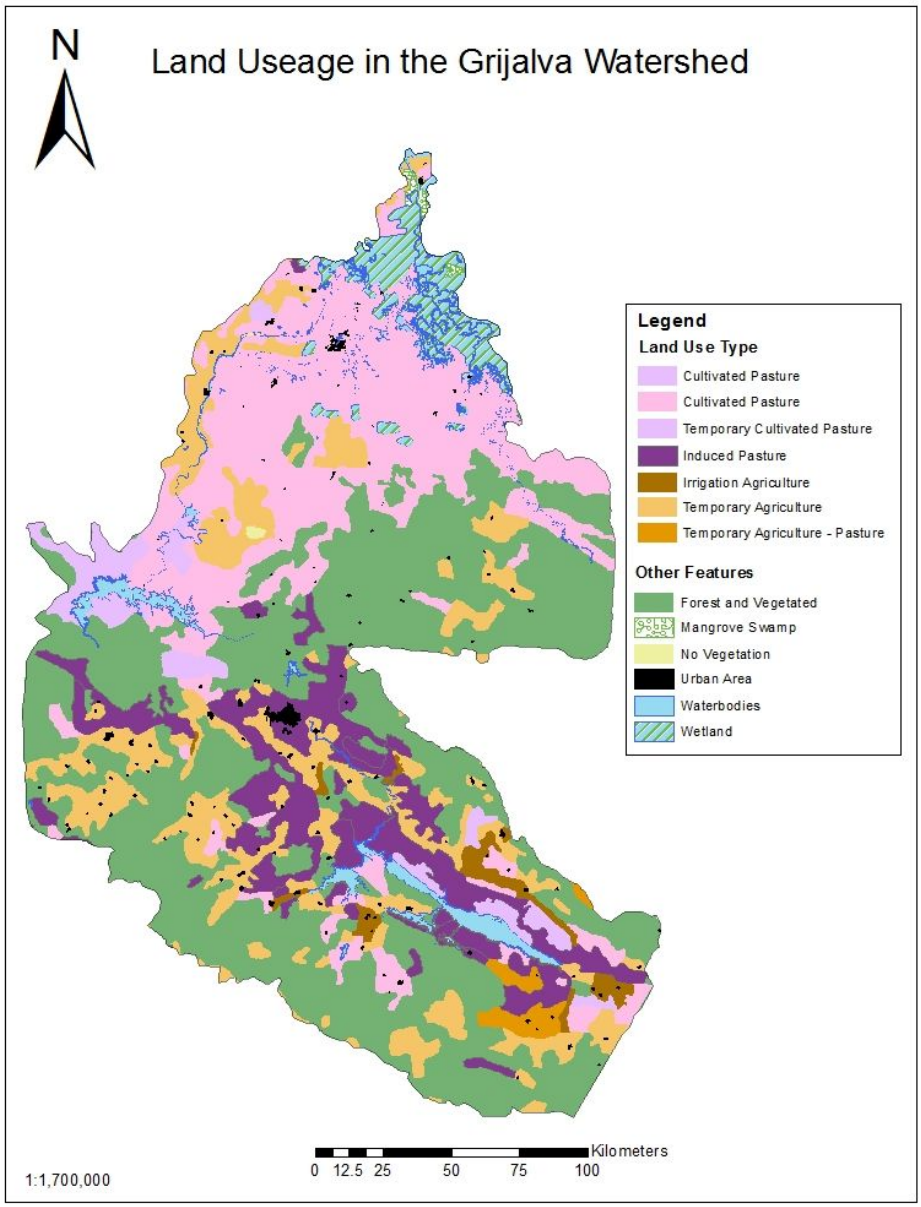

Figure 19 . Map of the current use of land within the watershed. A majority of the land is currently being used for agriculture and as pasture land.

6.5 Ecological Zones

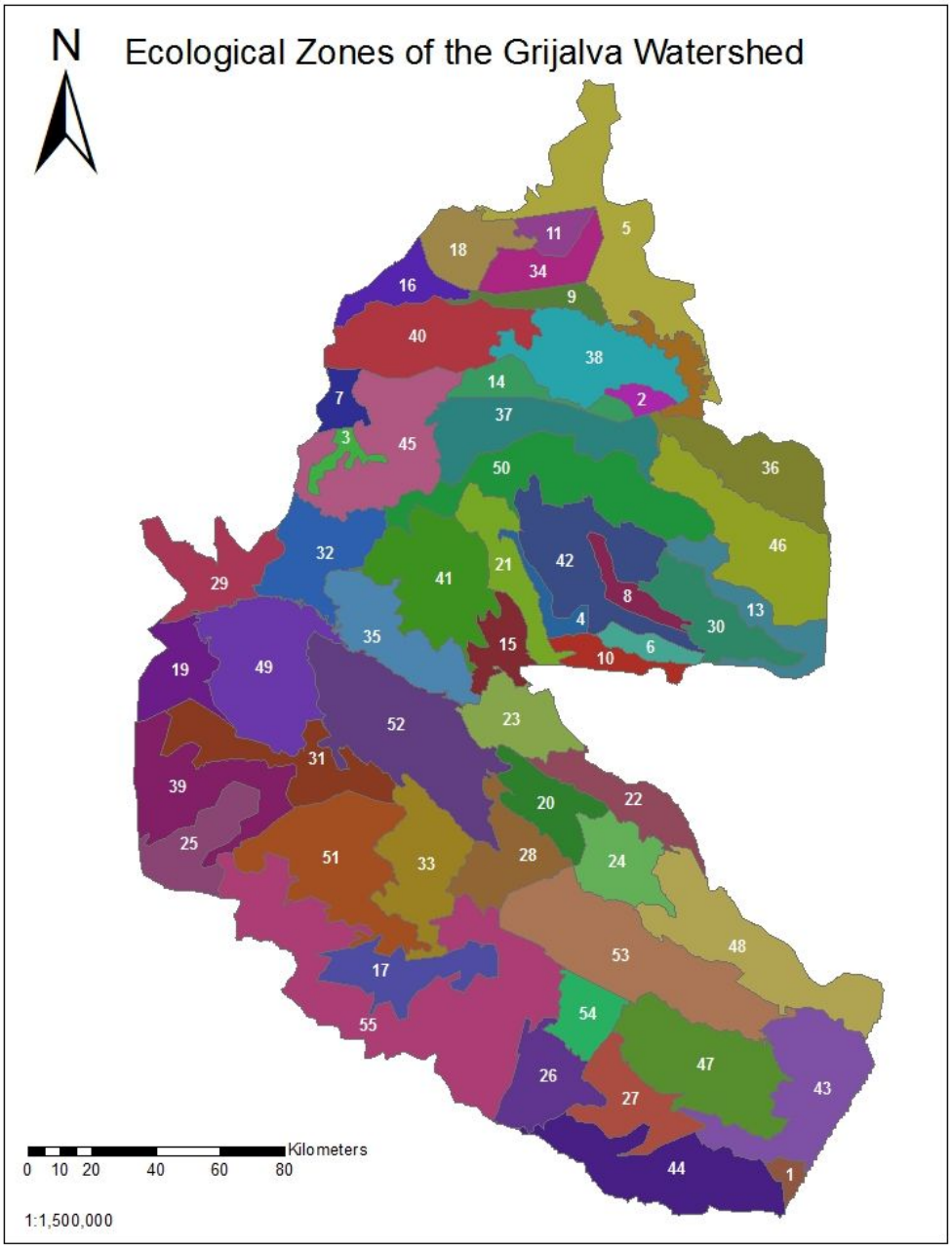

Figure 20. Shows the ecozones from the combined land resource data.

There are 55 ecozones total in the Grijalva watershed. These were determined by the intersection of the variables above (land use, geology, soil types, and vegetation) as well as the average temperatures and precipitation presented in section 6.1 of this report. Many of the environmental parameters contributing to the ecological zone delineation had to be generalized

to form more manageable polygons. This generalization reduces the resolution of spatial data, and creates less accurate representations out of necessity.

6.6 Watershed Land Capability, Suitability, and Land-Use Planning

6.6.1 Land Capability Assessment

Land capability is the general use that an area of land could support at maximum intensity without deterioration . Land capability was determined by extracting the attribute tables from the various ecological zones. These capability classes were designed by the FAO and range from I to VIII. The criteria for creating these classifications was based upon characteristics such as slope (topography), drainage (wetness), soil texture and depth (physical conditions) see classification table in appendix.

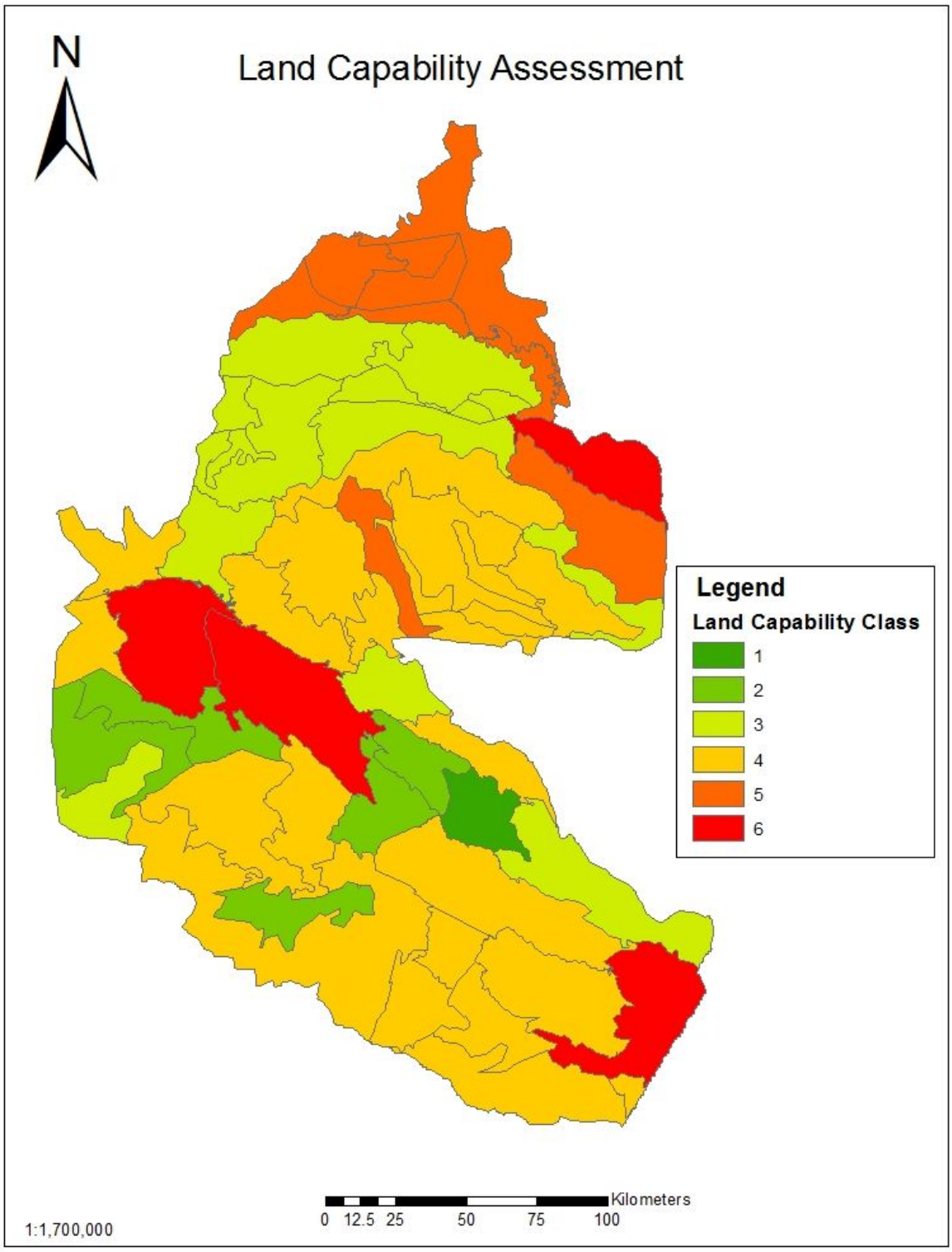

Figure 21. Land Capability is represented on a scale of 1 most capable, to 6 being least.

For the purposes of use-of-information in ArcGIS we associated numerical values to the roman numeral designations of the capability classes as defined by the FAO. Assessing our ecozones determined that we did not have any Class VII (7) or VIII (8) regions however many of our ecozones fell in the range of Classes III (3) to V (5) meaning that much of our landscape has a limited potential capability. The least capable areas are surrounding the large bodies of water in our watershed, Lake Angostura and the lake of the Malpaso dam.

Because of the physical characteristics associated with the FAO capability classes, it is likely that slope was a great factor in the area surrounding Lake Angostura because of its topography. Soil texture may have been the limiting factor in the central areas of our watershed.

6.6.2 Land Use Intensity Assessment

Land use intensity is a measure of the extent to which a land parcel is developed. It is important to determine this within watershed management because the intensity with which a region is being used can negatively impact the greater ecosystem. In the Grijalva Watershed land use intensity varies from minimal to great because there is such a variety of land use types.

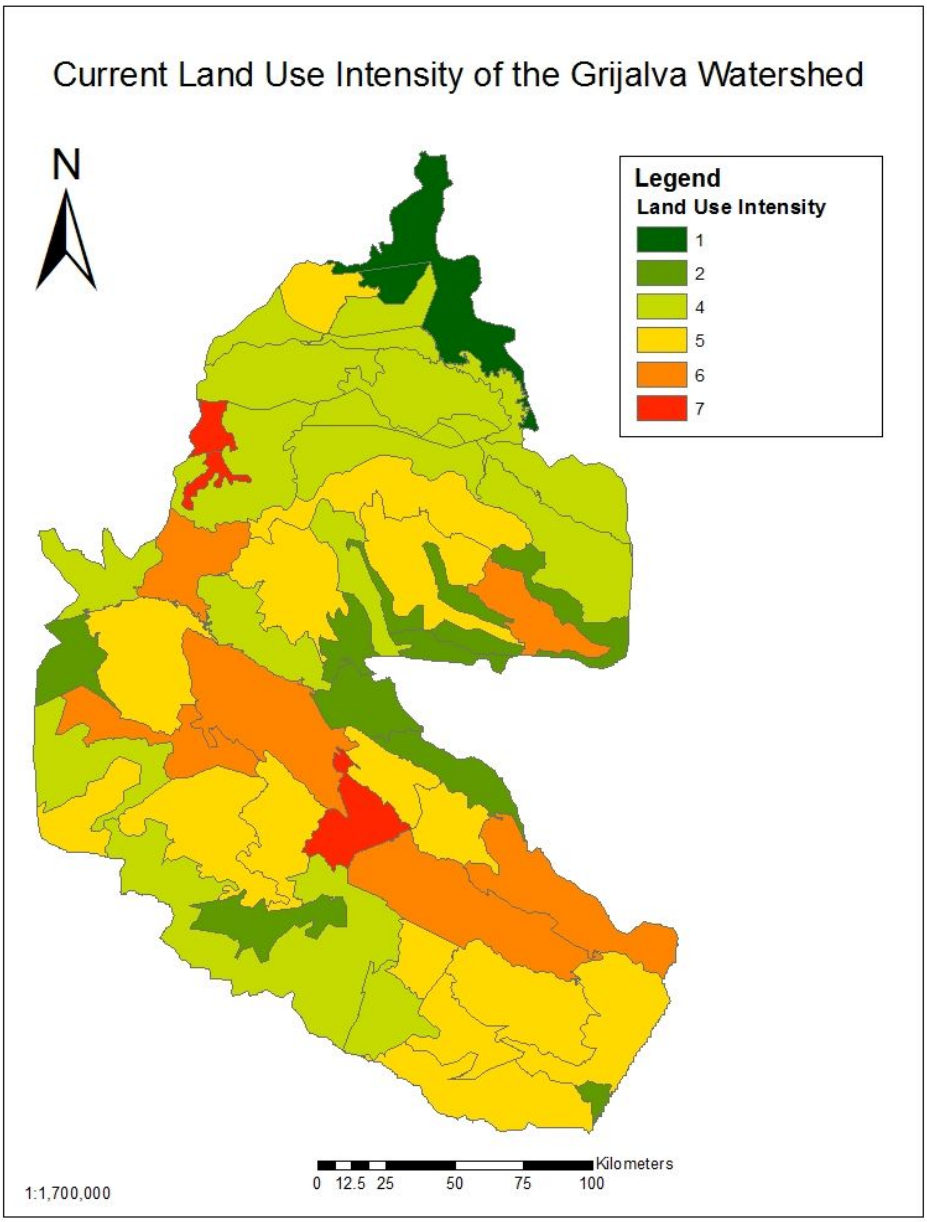

Figure 22. Land use intensity is represented on a scale of 1 as least intense and 9 as most intense.

Land use intensity was determined by categorizing the dominant land use types of each ecozone by the intensity classifications in the land capability assessment table. This table is available in Appendix_. We assigned a numerical value to each intensity (increasing from least intense to most) for ease of ArcGIS utilization. The values were associated as follows:

WILDLIFE = 1, FORESTRY = 2, LIMITED GRAZING = 3, MODERATE GRAZING = 4, INTENSE GRAZING = 5, LIMITED CULTIVATION = 6, MODERATE CULTIVATION = 7, INTENSE CULTIVATION = 8, and VERY INTENSE CULTIVATION = 9.

As shown in figure 22, the most intense land use occurs in the centre of our watershed, around Lake Angostura and near the Malpaso dam. This is of concern because of the proximity to large bodies of water. The wetlands at the Gulf coast were considered least intense because of dominant land use but it is worth mentioning that there is also agriculture occurring in these ecozones which could raise the land intensity significantly. Because of the nature of the agricultural practices in our watershed, intense cultivation and very intense cultivation were not present.

Where there were conflicting intensities between smaller land uses in individual ecozones, a conservative approach was taken. Visually determining an average intensity from the chart (as seen in the Appendix) helped to resolve and manage this.

6.8 Land Degradation Assessment

Land degradation assessments are essential to watershed management because they provide perspective on where land use is exceeding its physical capability. Using the capability assessment and intensity assessment together, areas of concern can be displayed as environmental hotspots. The areas should be considered as regions for which more suitable land use should be occurring. Depending on the degree of degradation these hotspots can change and be improved upon.

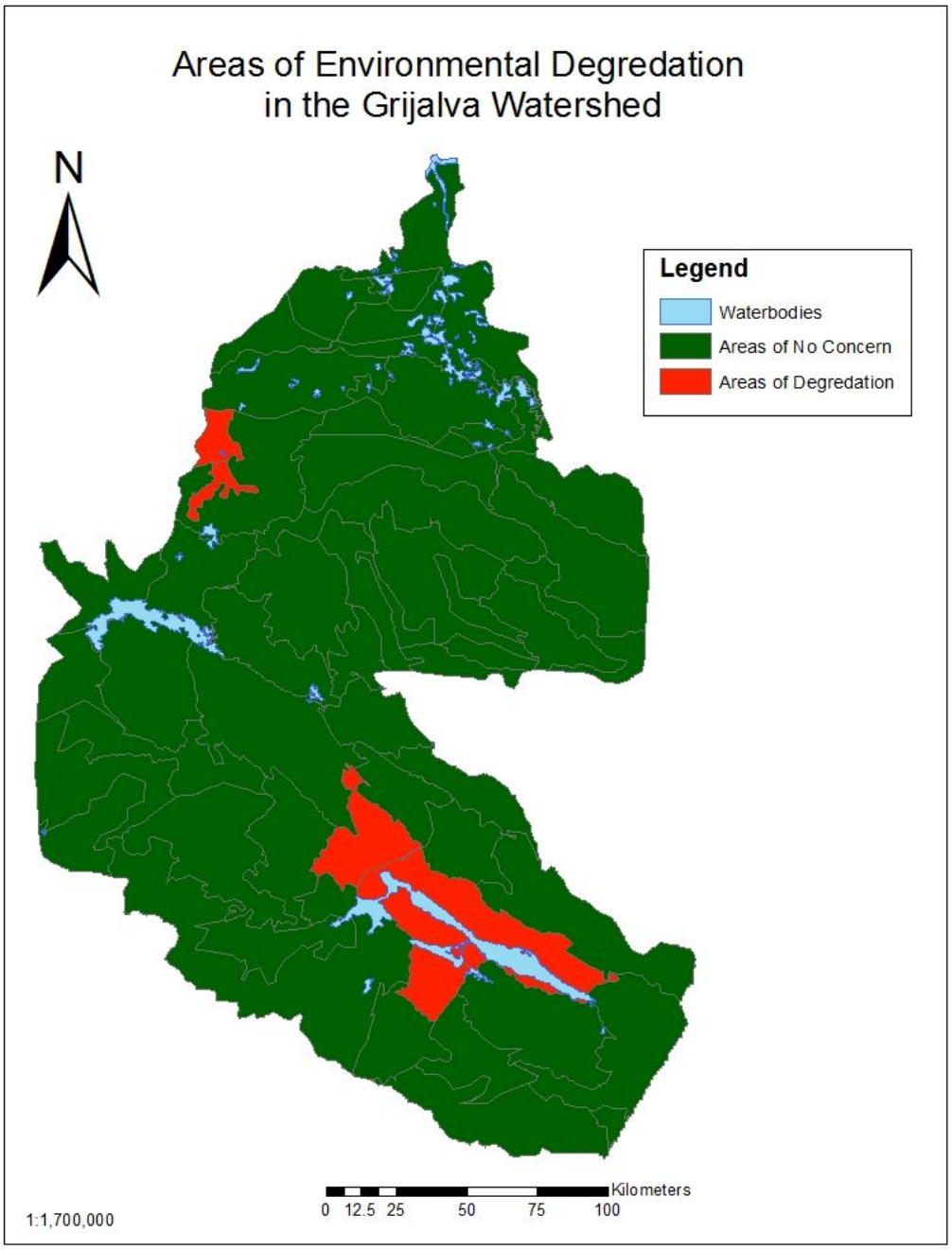

Figure 23. The above figure shows the areas of no concern (green) and the areas of degradation (red).

This degradation assessment was determined by comparing the capability class (figure 21) of each ecozone with its intensity designation (figure 22). A chart was used (as found in Appendix_) displaying land use intensities with appropriate capability classes. An environmental hotspot was formed wherever the land use intensity exceeded the capability of an ecozone as

shown in figure 23. The ecozones demonstrating environmental degradation coincide with the decreasing capability of land and increasing land use intensity.

The areas that are being degraded, both, fall under the category of agriculture. The area beside the La Angostura is mainly pastoral land indicating that there is over usage of this land resulting in erosion and sediment transport. This could be detrimental to the system since the area of concern is directly adjacent to and surrounding a major body of water. The runoff that is occurring in this area would be rich in nutrients and sediments because of the level of erosion that is occurring as well as the land use type. The second area of concern is located on the far northwestern side of the watershed. This area is being used as temporary agriculture. The fact that the area has been identified as an area of concern indicates that the characteristics of the land are not suitable for what it is being used for. The effects of land degradation include; a reduction in crop yields, and property damage from wind eroded soils (Scherr & Yadav, 1996)

6.9 Water Quantity and Quality in the Watershed

Water quality and quantity are excellent indicators of overall watershed health, and can provide researchers and watershed managers with invaluable insight into the ecological integrity of a system as well as an estimate of anthropogenic effects in the region.

Water quantity and quantity data was unfortunately unavailable in this region, and this limits the ability to integrate data into management plans.

6.9.1 Design of a Potential Monitoring Program for the Watershed

A monitoring program in the Grijalva watershed should focus upon three main elements of environmental characterisation i) water quality, ii) water quantity, and iii) soil chemistry;

<u>i) Water Quality</u>

Water quality refers to the physical, biological and chemical composition of water. Measuring the quality of water in the Grijalva would allow for adaptive management based on changes in these parameters and their likely causes within the watershed. Water chemistry can be characterised by measuring the water pH, conductivity, and alkalinity. The water can also be tested based on physical characteristics such as water turbidity and temperature and biological characteristics using indicators such as benthic macroinvertebrates.

<u>ii) Water Quantity</u>

Water quantity refers to the amount of water that runs through the watershed. This can be of interest and concern because the quantity of water in a watershed at any given time can contribute to flooding. In the Grijalva watershed this is of specific concern because of changing rainfall intensities, existing clay content in the soil, and the lowland areas near the gulf coast.

The steps for measuring water quantity in the watershed to predict flooding in the watershed could be as follows:

A. Measuring soil runoff data from soil morphology, which can be added to our maximum runoff, will make our runoff estimations more accurate
B. Climatological data from the graphs in Appendix A (graphs 1-17) show precipitation trends over the year. These can be refined for more watersheds and monitored in real time to predict and record precipitation across the watershed.
C. Combining the runoff with the precipitation data will allow us to monitor and predict flooding events downstream from the data collected upstream.

iii) Soil Chemistry

Soil chemistry can indicate the utility of soils as well as the resilience of some areas to pollution and a loss of nutrients. Organic matter content, cation exchange capacity, base saturation, particle size, and soil respiration should be assessed at numerous sites repeatedly to ensure there is not a depletion of soil nutrients or an acidification of systems.

7.0 - Discussion

Morphology: The watershed holds mountains in the South and leveled grasslands and wetlands at the North. The slope Elevation layer (Figure 18(b)) demonstrates the variability in the landscape.
The soil and geology layer show a similar pattern of uniformity in the North and diversity in the South of the watershed.
We used the FAO classification as is used for data collection by the Mexican government and its partner institutions.
The stream system was dotted with Vernal streams, showing a significant impact of the monsoon season on the hydrological landscape.
Various efforts have been seen to manage water with several dams dotting the stream sections and several artificial canals running through agricultural communities.
The watershed also houses at least 3 distinct water bodies that are fed by the main stream system.

Climatology: The watershed is heavily dependant on rainfall. The temperature scales are mostly growable, except the mountainous region on the Southern coast of the watershed. Where minimum temperatures can fall below freezing during the peak of winter. The average temperature however, for the entire watershed, maintains above the 6.5 degree Centigrade, that we use as a threshold. There is however, strict dependence of the watershed on rainfall, which is quite variable and lowest at the bottom of the watershed. It is also the governing factor for the Length of Growing Period in the watershed.

Runoff: The Grijalva river has wide large runoff areas associated with almost all but 1 watershed. This comes primarily from the branching of the Grijalva River, rendering it a large catchment area from the whole watershed. Slopes in the landscape increase the surface area that

that section can offer. Considering the previously mentioned precipitation values and this observed runoff estimation, we see significant potential for flooding in the region, especially due to the proximity of stream channels.

Land Utilization: The marked 'hotspots', are ecozones which were exceeding their capability, a total of 4 in total. Majority of the ecozones demonstrated either appropriate or less than capacity use intensity. However, these ecozones are aggregation of smaller intersects, and each ecozone can house smaller section(s) of over-utilization. The ecozones classified as hotspots however, show an aggregation of overutilization and are areas of concern.

Due to the size of the Grijalva watershed, it has 11 distinct sub watersheds **(Figure 2)**, and also consists of 55 ecozones **(Figure 20)**. As a result of this, the watershed shows a high variability in data. For example, the geology of the watershed is complex. **Figure 9** presents the geology of the watershed, It is evident that a majority of the southern regions made up of igneous materials, a rock formation that occurs with the solidification of lava. This is plausible, considering the watershed is located on an ancient volcanic zone. A majority of the northern watershed consists of caliza (limestone), which forms in marine environments with the decay of organic materials such as shells.

The geology of the region has a direct influence on the soils of the region. **Figure 10** defines the soil classification of the Grijalva Watershed. The southern region of the watershed consists of the soil classification vertisol. Vertisol consists mainly of clay, a feature that is present in areas that contain igneous parent material. The northern region showed high concentrations of acrisol, which is also high in clay. Soils ultimately determine what can grow and what can't. **Figure 11** represents the type of vegetation present in the watershed. The watershed shows high variability in types of vegetation. The northern region, in which the discharge point is located at, is mainly comprised of wetlands. A majority of the region consists of pasture, which is home to many livestock animals such as cattle. The central part of the region contains a jungle with coniferous and deciduous vegetation. The southern region consists of pine-oak forest, no vegetation, and agriculture, the biggest contributor to the local economy.

Due to the type of vegetation, it has influenced the watershed regions land usage, **Figure 19** defines the results for land usage in the Grijalva Watershed. It is evident that most of the watershed consists of cultivated pastures, and forestry. There are pockets of urban areas, which are mainly situated in or near agriculture and pastures. As mentioned earlier, the north-most region consists of wetlands. Wetlands are a big sector for the fishing industry, and is threatened due to increased sea level. With Sea level rising, the wetlands and areas in close proximity will be submerged under water within the next century.

The Grijalva Watershed is very mountainous to the south, which has led to the distinct morphology found in the region. **Figure 18(b)** consists of the slope for the region, in which the area to the north has no slope, and the area to the south has a slope range from 10.60-32.883 degrees. This has a direct impact on runoff, as slope is a major determining factor. The regions with the highest degrees of slope, also showed a correlation with the highest runoff. This is accurately represented in **Figure 18(a).** The study also found that areas with high runoff rates

are also more prone to environmental degradation. This is evident in **Figure 23**, in which erosion is a major concern for the reason. With increased severe weather events to occur more frequently, it is predicted that this red range will continue to grow.

8.0 - Bibliography

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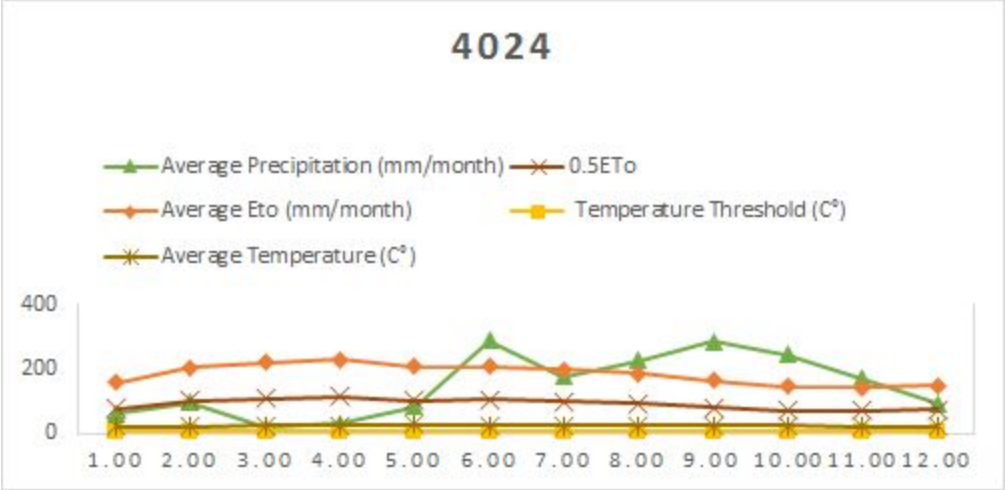
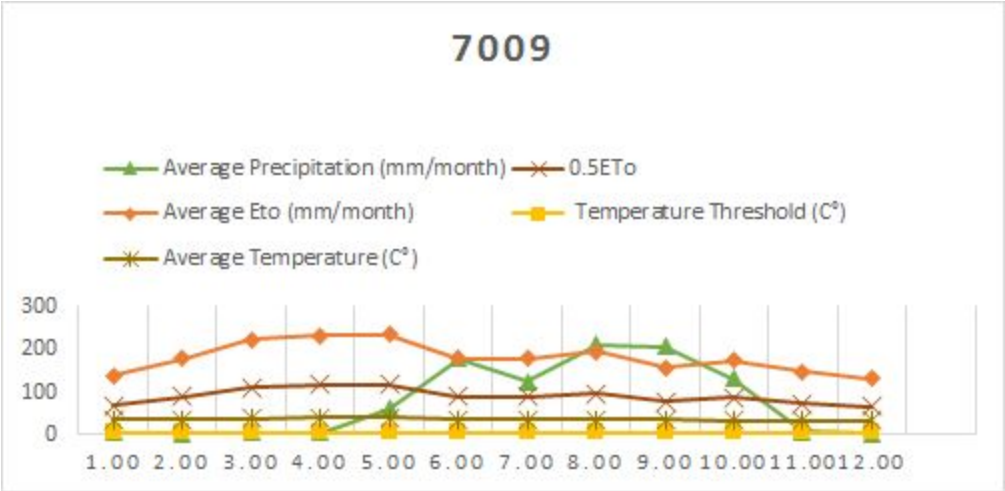
9.0 List of Appendices

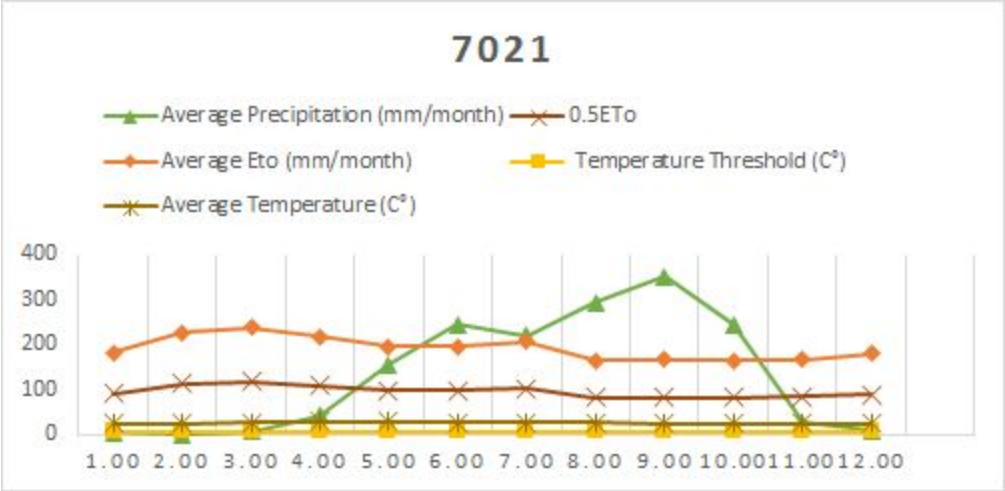
Table A.1: List of weather stations used.

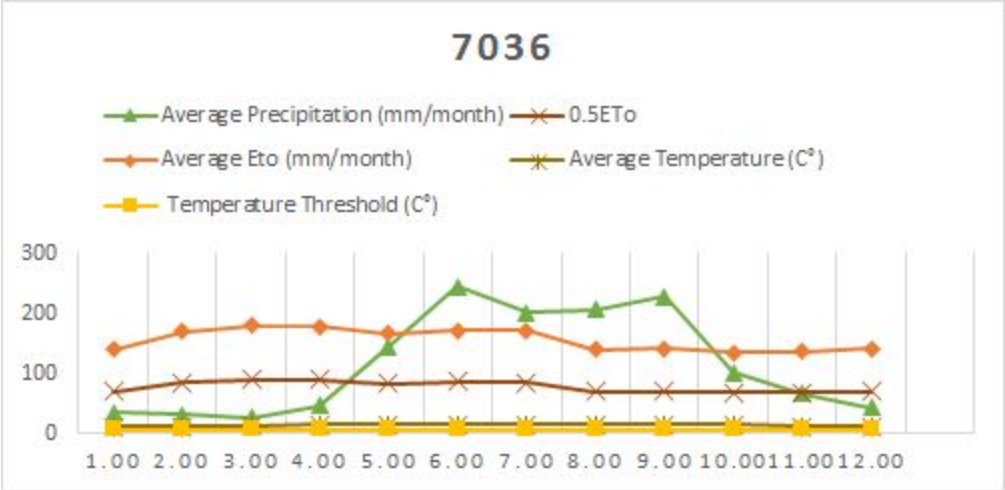
#	NAME	ID	ALTITUDE (M)	LATITUDE	LONGITUDE
1	PALIZADA	4024	4	18.253056	-92.091944
2	AQUESPALA	7009	617	15.794167	-91.920278
3	CATARINITAS	7021	945	15.9025	-92.482778
4	CHILIL	7036	2266	16.677778	-92.489167
5	FINCA A. PRUSIA	7054	1040	15.4355	-92.4739
6	OXOLOTAN	27070	63	17.3806	-92.75
7	GUADALUPE GRIJALVA	7070	630	15.693611	-92.161111
8	GUAQUIPTEPEC	7071	1160	17.144722	-92.290278
9	LA MESILLA	7094	1210	16.184167	-92.2875
10	BUENOS AIRES	7333	1820	15.3325	-92.2675
11	MONTERREY	7349	700	16.058889	-93.368889
12	FRANCISCO I. MADERO	7376	736	16.802778	-93.755278
13	TRES BRAZOS	27050	2	18.3842	-92.6
14	MACUSAPANA	27030	10	17.75	-92.6
15	TAPIJULAPA	27042	44	17.461111	-92.7775
16	TANCOCHAPA	30167	13	17.8736	-94.0889

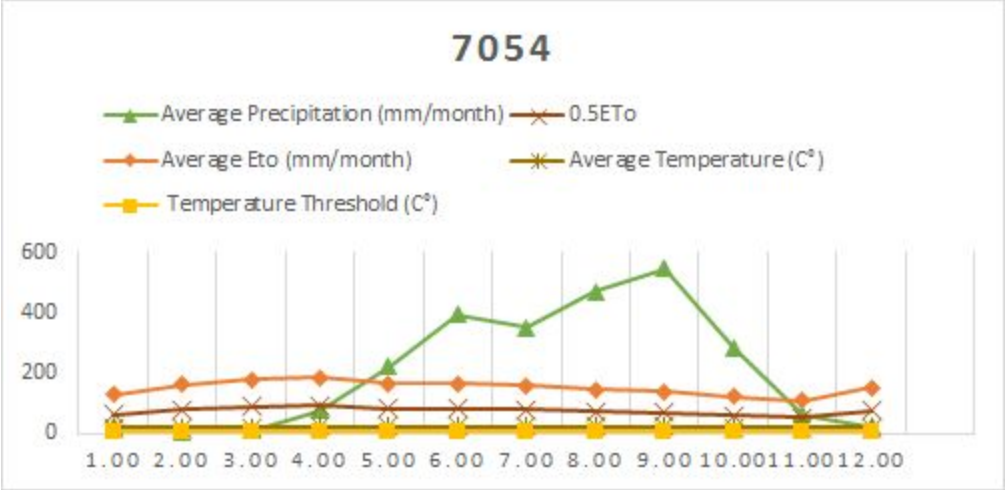
***Graphs (1 to 17) showing the Annual climatology for the weather stations.**

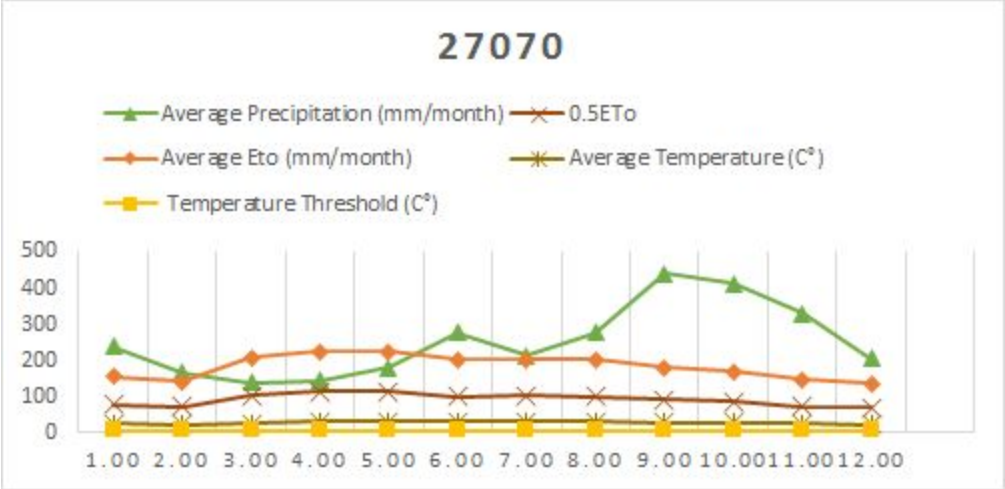


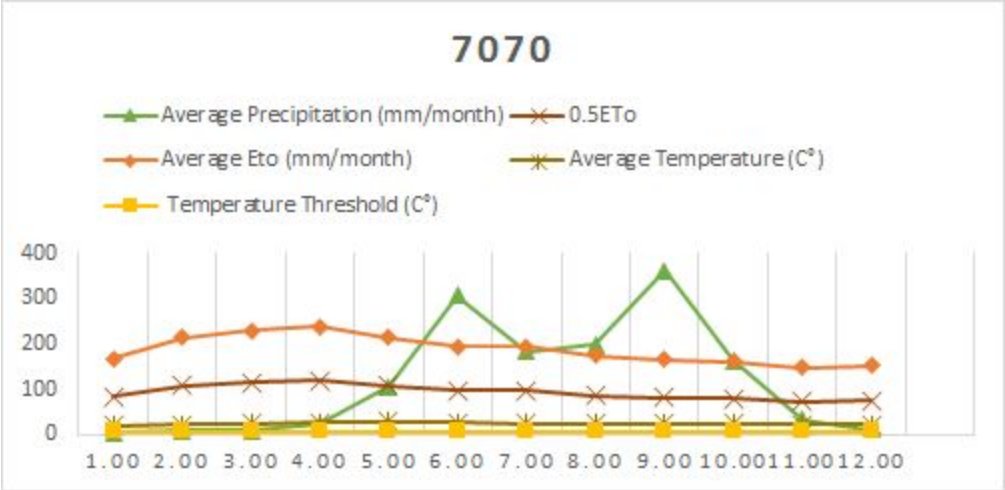


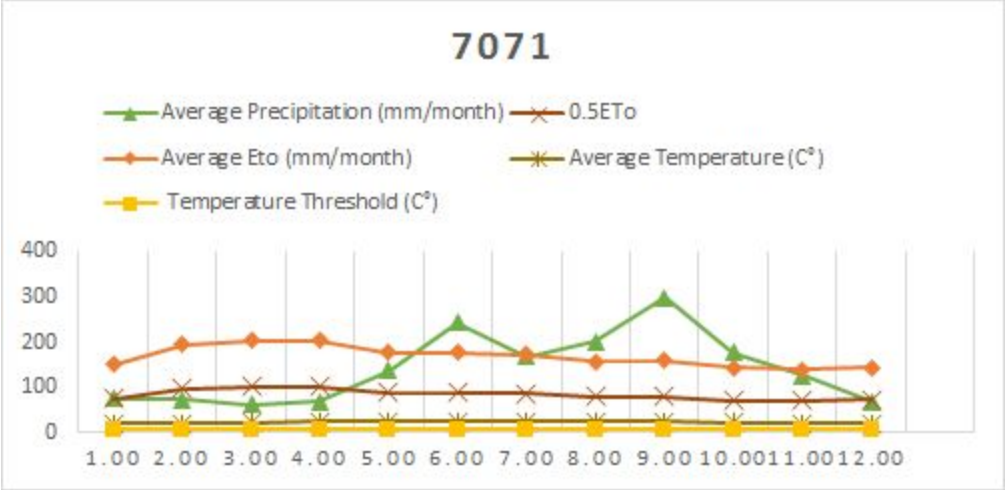


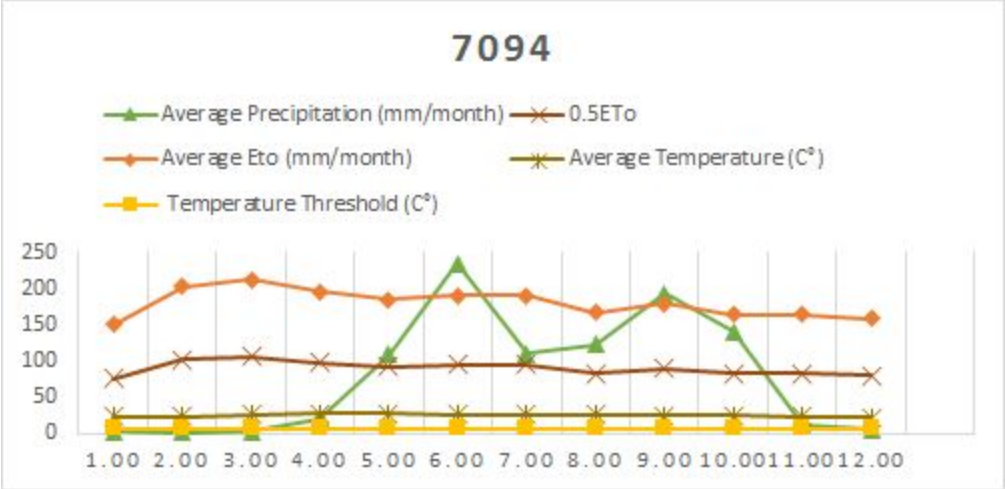

7036
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
300
200
100
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

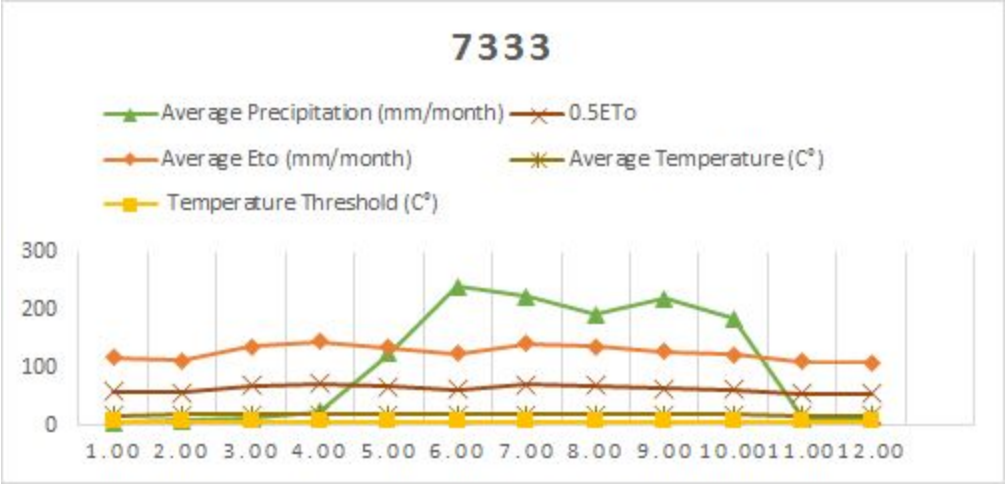

7054
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
600
400
200
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

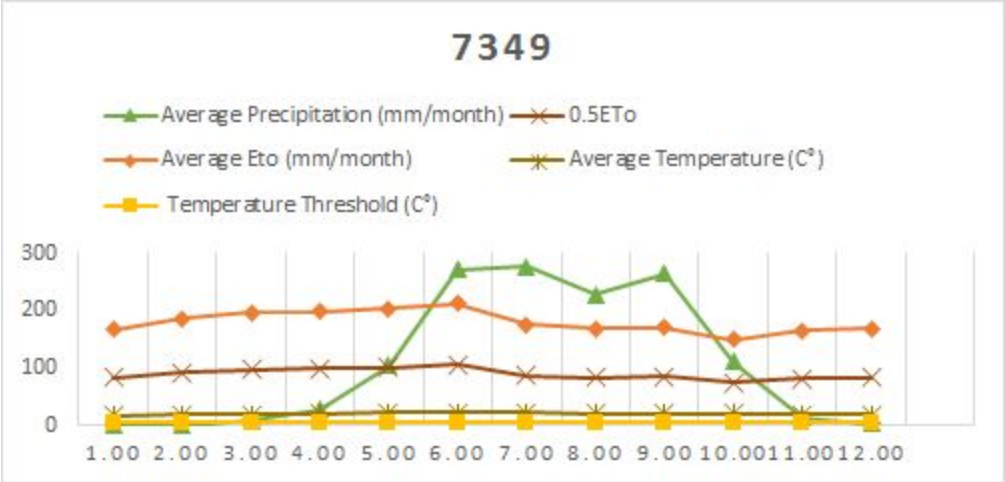

27070
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
500
400
300
200
100
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

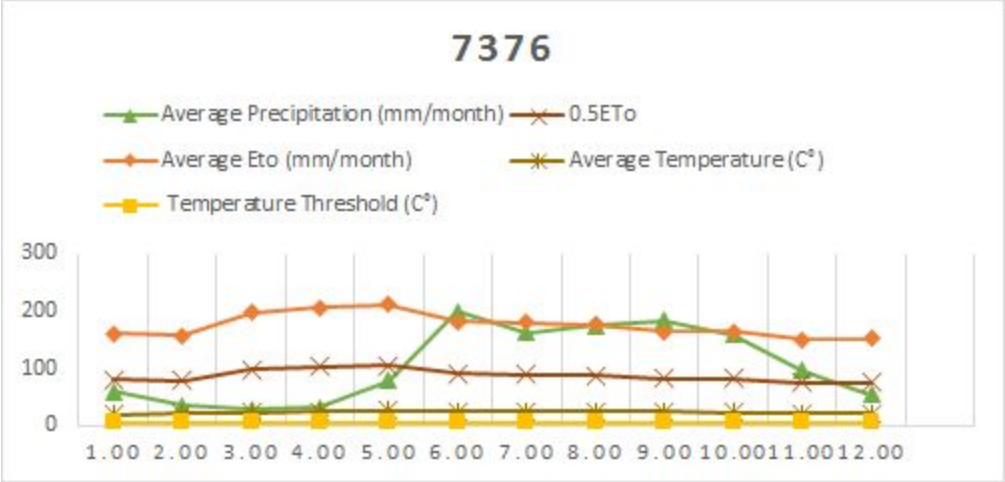

7070
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
400
300
200
100
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

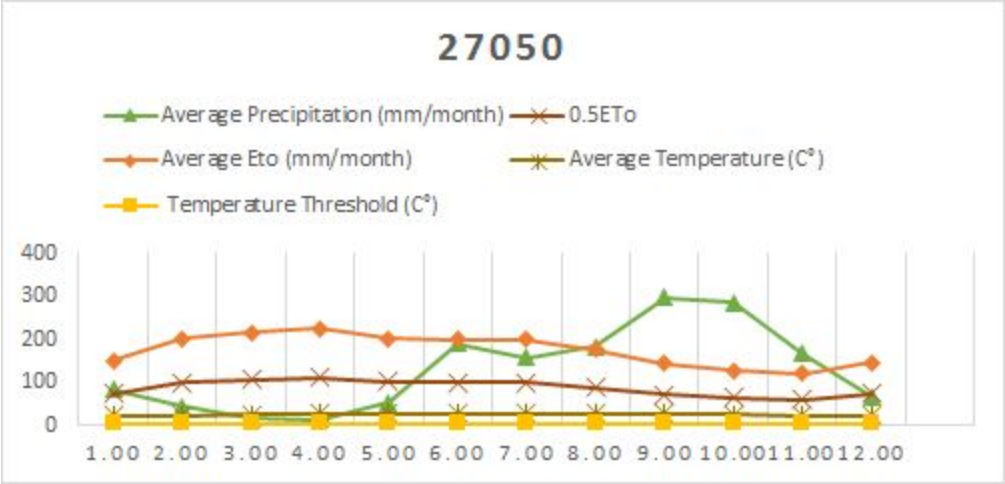

7071
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
400
300
200
100
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

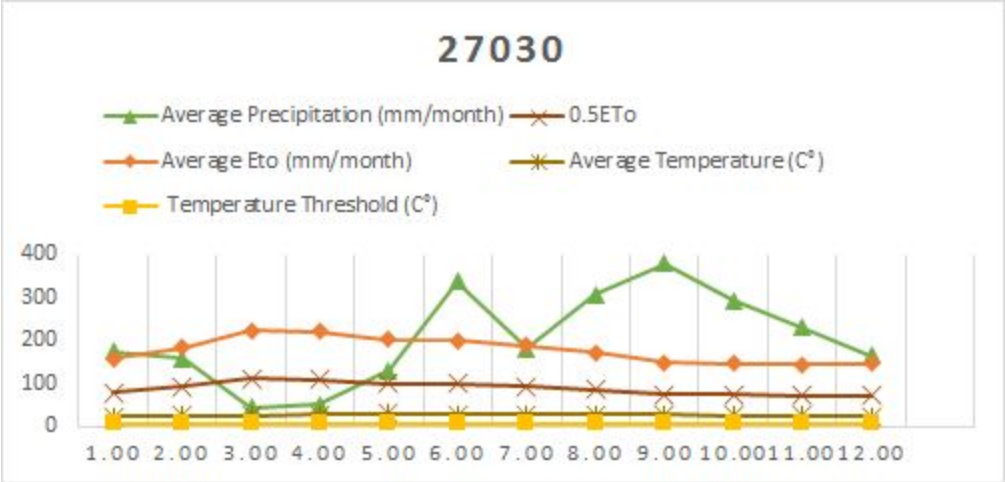

7094
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
250
200
150
100
50
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

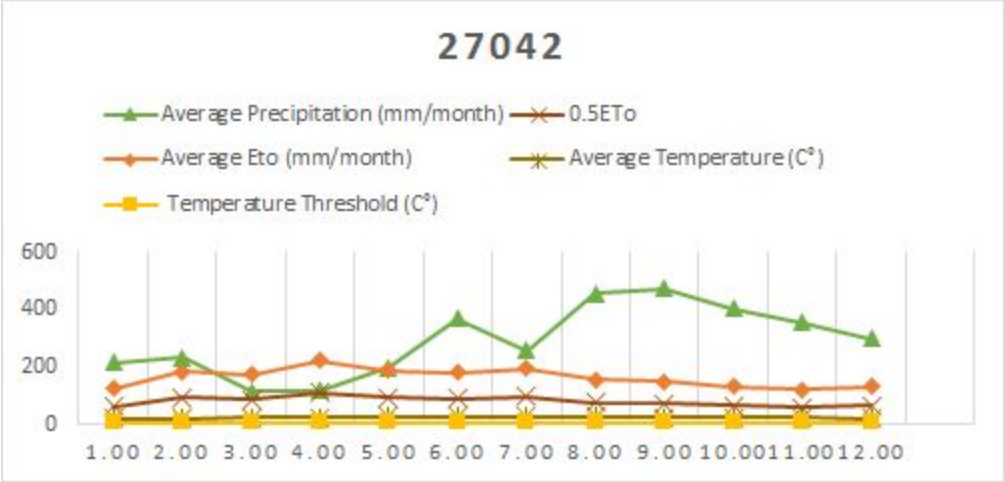

7333
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
300
200
100
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

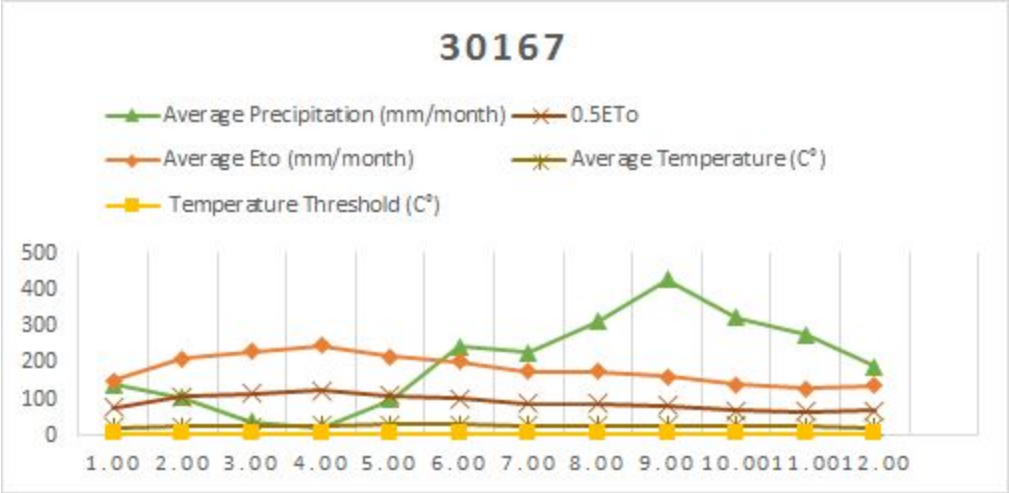

7349
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
300
200
100
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00


7376
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
300
200
100
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

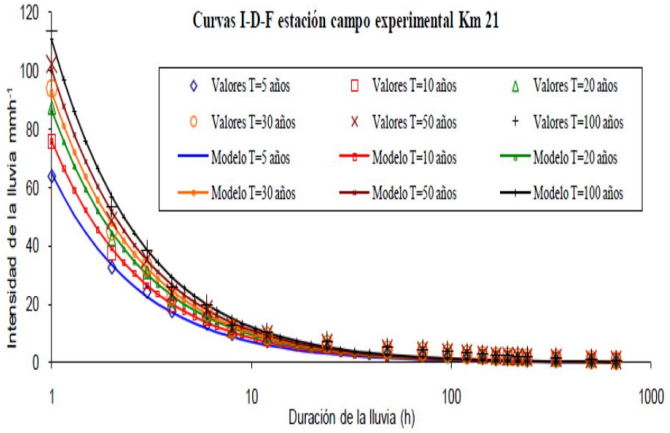

27050
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
400
300
200
100
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

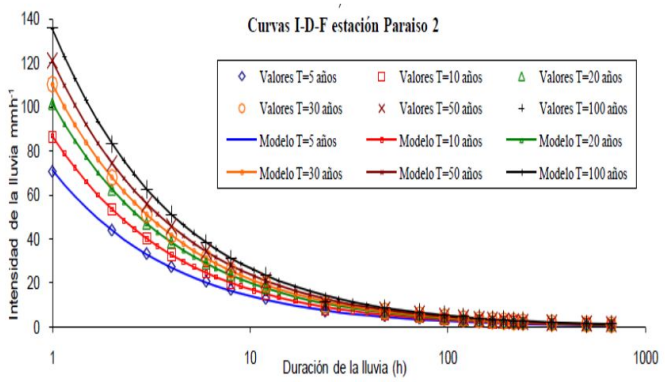

27030
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
400
300
200
100
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

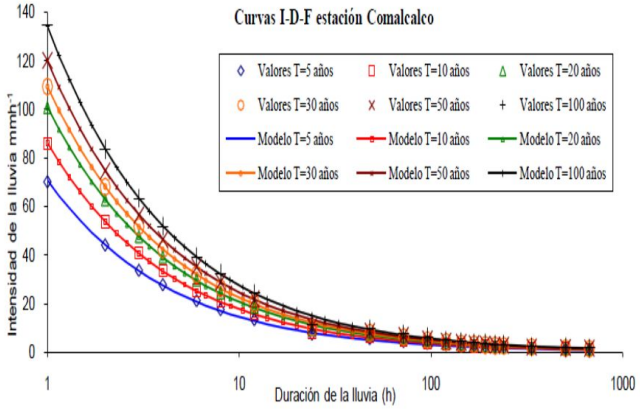

27042
Average Precipitation (mm/month)
0.5ETo
Average Eto (mm/month)
Average Temperature (C°)
Temperature Threshold (C°)
600
400
200
0
1.00 2.00 3.00 4.00 5.00 6.00 7.00 8.00 9.00 10.00 11.00 12.00

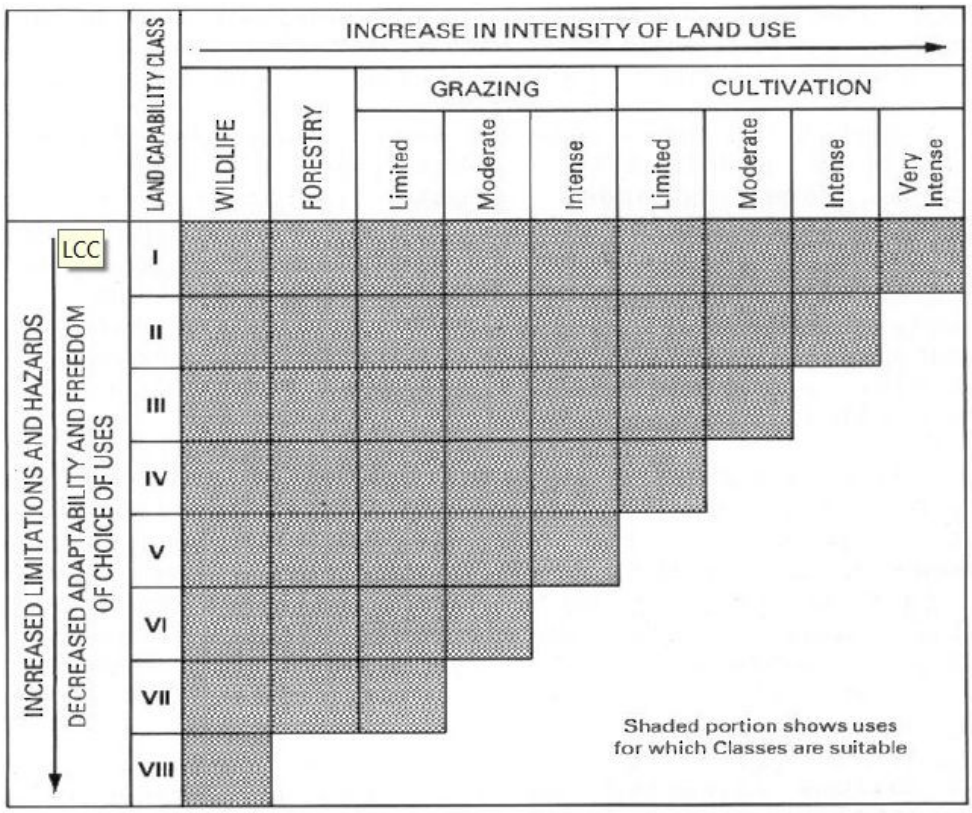


***IDF Curves Grijalva Mexico 1-3**


Curvas I-D-F estación campo experimental Km 21
Valores T=5 años
Valores T=10 años
Valores T=20 años
Valores T=30 años
Valores T=50 años
Valores T=100 años
Modelo T=5 años
Modelo T=10 años
Modelo T=20 años
Modelo T=30 años
Modelo T=50 años
Modelo T=100 años
Intensidad de la lluvia mmh-1
Duración de la lluvia (h)
0
20
40
60
80
100
120
1
10
100
1000


Curvas I-D-F estación Paraiso 2
Valores T=5 años
Valores T=10 años
Valores T=20 años
Valores T=30 años
Valores T=50 años
Valores T=100 años
Modelo T=5 años
Modelo T=10 años
Modelo T=20 años
Modelo T=30 años
Modelo T=50 años
Modelo T=100 años
Intensidad de la lluvia mmh-1
Duración de la lluvia (h)
0
20
40
60
80
100
120
140
1
10
100
1000


Curvas I-D-F estación Comalcalco
Valores T=5 años
Valores T=10 años
Valores T=20 años
Valores T=30 años
Valores T=50 años
Valores T=100 años
Modelo T=5 años
Modelo T=10 años
Modelo T=20 años
Modelo T=30 años
Modelo T=50 años
Modelo T=100 años
Intensidad de la lluvia mmh-1
Duración de la lluvia (h)
0
20
40
60
80
100
120
140
1
10
100
1000



Characteristic	Class I	Class II	Class III	Class IV	Class V	Class VI	Class VII	Class VIII
Topography (t)								
Slope(%)	< 2	< 6	< 12	< 25	< 25	< 25	< 55	> 55
Wetness (w)								
Flooding	No (F_o)	No (F_o)	No (F_o)	No to Slight (F_1)	No to Severe (F_3)	No to Severe (F_3)	No to Very Sever (F_4)	No to Very Sever (F_4)
Drainage	Good	Moderate or better	Somewhat imperfect or better	Imperfect or better	Poor or Better	Poor or Better	Very poor or Better	Very poor or Better
Physical Soil Conditions (s)								
Surface Texture	SL to Co	Lfs to C-60s	Fs-C-60v	cS to C+60b	cS to Om	cS to Om	cS to Om	cS to Om
Surface coarse fragments (Vol %)	None	< 15	< 35	< 55	< 55	< 55	< 75	> 75
Surface Stoniness (%)	None	< 0.01	< 0.1	< 0.3	< 15	< 15	< 75	> 75
Rockiness (%)	None	< 2	< 10	< 25	< 50	< 50	< 75	> 75
Subsurface texture	L to C-60s	SCL to C-60v	LfS to C+60v	fS to C+60v	cS to Om	cS to Om	cS to Om	cS to Om
Subsurface coarse fragments	< 15	< 35	< 55	< 75	< 75	< 75	> 75	> 75
Soil Depth (m)	> 1.5	> 1	> 0.50	> 0.25	> 0.25	> 0.25	> 0.10	< 0.10